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Satellite Earth Stations and Systems (SES);
Vehicle-Mounted Earth Stations (VMES)
operating in the 14/12 GHz frequency bands;
Harmonised Standard for access to radio spectrum

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Foreword

This draft Harmonised European Standard (EN) has been produced by ETSI Technical Committee Satellite Earth Stations and Systems (SES), and is now submitted for the combined Public Enquiry and Vote phase of the ETSI Standardisation Request deliverable Approval Procedure (SRdAP).

The present document has been prepared under the Commission's standardisation request C(2015) 5376 final [i.5] to provide one voluntary means of conforming to the essential requirements of Directive 2014/53/EU on the harmonisation of the laws of the Member States relating to the making available on the market of radio equipment and repealing Directive 1999/5/EC [i.6].

Once the present document is cited in the Official Journal of the European Union under that Directive, compliance with the normative clauses of the present document given in Table A.1 confers, within the limits of the scope of the present document, a presumption of conformity with the corresponding essential requirements of that Directive and associated EFTA regulations.

Proposed national transposition dates	
Date of latest announcement of this EN (doa):	3 months after ETSI publication
Date of latest publication of new National Standard or endorsement of this EN (dop/e):	6 months after doa
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Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document specifies technical characteristics and methods of measurements for Vehicle-Mounted Earth Stations (VMES). The VMES system overview is presented below.

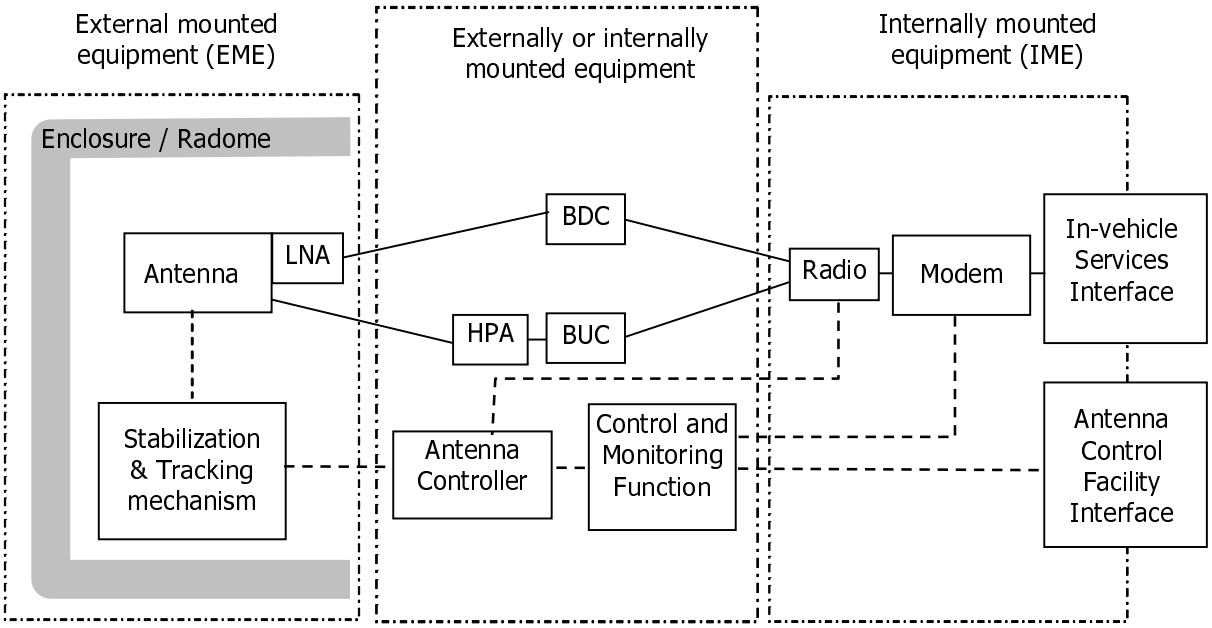


Figure 1: VMES System Overview

- The VMES may transmit and receive data when the vehicle is in motion and also when the vehicle is stationary.
- The VMES operates on wheeled or tracked vehicles and, therefore, may be subject to occasional disturbances and interruptions in the satellite link.
- The VMES is operating as part of a satellite network (e.g. star, mesh or point-to-point) used for the distribution and/or exchange of information.
- The VMES is comprised of all the equipment, electrical and mechanical, from the antenna itself to the interface with other communications equipment on a vehicle (usually referred to as the terrestrial interface).
- The VMES transmits on single carrier in the frequency range 14,00 GHz to 14,50 GHz.

NOTE 1: For the purposes of the present specification, OFDM modulation is considered as a single carrier.

- The VMES receives in one or more frequencies within the range from 10,70 GHz to 12,75 GHz.
- The VMES uses linear or circular polarization.
- The VMES is designed to operate through a geostationary satellite (or a cluster of co-located geostationary satellites) that is at least 3° away from any other geostationary satellite operating in the same frequencies and over the same coverage area.

NOTE 2: Satellites may be spaced closer than 3°. In such cases, the satellite operator will inform the VMES client of the requirements of the system coordination agreements.

- The VMES transmits at elevations greater than or equal to 7° relative to the local horizon.
- The VMES is designed for unattended operation.

- The VMES is designed for both mobile and stationary operation. In the case of stationary operation, the VMES should not be accessible to the general public and operated safely.
- The VMES is controlled and monitored by an Antenna Control Facility (ACF). This function may be performed centrally (e.g. for a network of VMESs with a central hub) or it could be performed within the VMES for autonomous control. The ACF is outside the scope of the present document.

The present document applies to the VMES with its ancillary equipment and its various telecommunication ports, and when operated within the boundary limits of the operational environmental profile as defined for its intended use and when installed as required by its intended use or in the user documentation.

NOTE 3: The relationship between the present document and essential requirements of article 3 of Directive 2014/53/EU 2 [i.6] is given in Annex A.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

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The following referenced documents are necessary for the application of the present document.

[1] Void.

[2] CISPR 16-1-1:2019: "Specification for radio disturbance and immunity measuring apparatus and methods - Part 1-1: Radio disturbance and immunity measuring apparatus - Measuring apparatus".

[3] CISPR 16-1-5:2014: "Specification for radio disturbance and immunity measuring apparatus and methods - Part 1-5: Radio disturbance and immunity measuring apparatus - Antenna calibration sites and reference test sites for 5 MHz to 18 GHz".

[4] EN 60529:1991+A1:2000+A2:2013+AC:2016-12: "Degrees of protection provided by enclosures (IP code)", (produced by CENELEC).

[5] EN 60068-2-6:2008: "Environmental testing - Part 2-6: Tests - Test Fc: Vibration (sinusoidal)", (produced by CENELEC).

[6] EN 60068-2-11:2021: "Environmental testing - Part 2-11: Tests - Test Ka: Salt mist", (produced by CENELEC).

[7] EN 60068-2-27:2009: "Environmental testing - Part 2-27: Tests - Test Ea and guidance: Shock", (produced by CENELEC).

2.2 Informative references

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[i.1] Void.

[i.2] Void.

[i.3] ETSI TR 102 375: "Satellite Earth Stations and Systems (SES); Guidelines for determining the parts of satellite earth station antenna radiation patterns concerned by the geostationary satellite orbit protection".

[i.4] ETSI TR 102 215: "Electromagnetic compatibility and Radio spectrum Matters (ERM); Recommended approach, and possible limits for measurement uncertainty for the measurement of radiated electromagnetic fields above 1 GHz".

[i.5] Commission implementing decision C(2015) 5376 final of 4.8.2015 on a standardisation request to the European Committee for Electrotechnical Standardisation and to the European Telecommunications Standards Institute as regards radio equipment in support of Directive 2014/53/EU of the European Parliament and of the Council.

[i.6] Directive 2014/53/EU of the European Parliament and of the Council of 16 April 2014 on the harmonisation of the laws of the Member States relating to the making available on the market of radio equipment and repealing Directive 1999/5/EC (RE Directive).

[i.7] MIL STD 810F (2000): "DOD Test Method Standard for Environmental Engineering Considerations and Laboratory Tests".

[i.8] ETSI EG 203 336: "Guide for the selection of technical parameters for the production of Harmonised Standards covering article 3.1(b) and article 3.2 of Directive 2014/53/EU".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in Directive 2014/53/EU [i.6] and the following apply:

ancillary equipment: equipment used in connection with a VMES is considered as ancillary if the three following conditions are met:

- the equipment is intended for use in conjunction with the VMES to provide additional operational and/or control features (e.g. to extend control to another position or location); and
- the equipment cannot be used on a stand-alone basis, to provide user functions independently of the VMES; and
- the absence of the equipment does not inhibit the operation of the VMES

Antenna Control Facility (ACF): set of functional entities that, at system level, monitor and control the correct operation of the VMES and, if appropriate, all of the VMESs in a network

antenna controller: equipment used to maintain antenna stabilization and tracking accuracy based on inputs from the Control and Monitoring Function

carrier-off radio state: radio state in which the VMES may transmit and does not transmit any carrier

NOTE 1: The phrase "the VMES may transmit" means that all the conditions for transmission are satisfied (e.g. in a state where transmissions are permitted, no failure detected, and the VMES is correctly pointed towards the satellite).

NOTE 2: The existence of a "Carrier-off" radio state depends on the system of transmission used. For VMESs designed for continuous transmission mode there may be no "Carrier-off" state.

carrier-on radio state: radio state in which the VMES may transmit and transmits a carrier

Control Channel (CC): channel or channels by which VMESs receive control information from the ACF

$EIRP_{max}$: maximum e.i.r.p. capability of the VMES as defined for the intended use of the VMES

emissions-disabled radio state: radio state in which the VMES does not emit

NOTE: Examples of cases where the VMES is in this radio state: before system monitoring pass, before the control channel is received, when a failure is detected, when a VMES is commanded to disable, and when the VMES is in a location requiring cessation of emissions.

external control channel: control channel which is either:

i) carried by the VMES network via the same or another satellite, but not within the internal protocol of the VMES system; or

ii) carried by any other radio communication system

external response channel: response channel which is either:

i) carried by the VMES network via the same or another satellite, but not within the internal protocol of the VMES system; or

ii) carried by any other radio communication system

Externally Mounted Equipment (EME): that part of the VMES intended to be installed on the outside of the vehicle (usually the roof), as defined for the intended use of the VMES, or as indicated in the user documentation

NOTE 1: The EME unit is usually comprised of the following main parts:

a) The antenna sub-system which converts the incident radiation field into a guided wave and vice versa.

b) The Low Noise Amplifier (LNA) which is a device that amplifies, with very low internal noise, the received signals in the Radio Frequency (RF) band.

c) The stabilization and tracking mechanics that ensure pointing of the antenna main beam towards the satellite within the required accuracy.

d) The radome for protecting the antenna subsystem, if required for normal operation.

NOTE 2: The LNA could be replaced by an LNB, which includes the block down converter.
Optionally, the EME may include the following elements (otherwise they will be part of the IME):

a) A Block Up Converter (BUC) and High Power Amplifier (HPA) which convert from the intermediate frequency to RF and amplify the low level RF signals for transmission through the antenna subsystem.

b) The Antenna Controller Processor.

c) The Control and Monitoring Function.

NOTE 3: The installation equipment (means of attachment) is outside the scope of the present document. However, the antenna structures and other components directly mounted on the antenna and forming an integral part of it, are subject to the specifications of the present document.

NOTE 4: Certain configurations may include more than one EME per VMES. As for example, the VMES may have separate transmit and receive antennas or it may have redundant transmit/receive antenna units.

integral antenna: antenna which may not be removed during the tests according to EUT documentation

internal control channel: control channel which is carried by the VMES network via the same satellite as used for transmission of user data and within the internal protocol structure of the VMES system

internal response channel: response channel which is carried by the VMES network via the same satellite as used for transmission of user data and within the internal protocol structure of the VMES system

Internally Mounted Equipment (IME): part of the VMES equipment which is installed inside the vehicle and its connection cables with the EME

NOTE: The IME is usually comprised of:

a) the modem and the IF radio equipment;

b) the interfaces to equipment and services onboard the vehicle.

Optionally the IME may include the following elements (otherwise they will be included in the EME):

c) a Block Down Converter (BDC), which converts the RF signals to intermediate frequencies;

d) a Block Up Converter (BUC) and High Power Amplifier (HPA) which convert from the intermediate frequency to RF and amplify the low level RF signals for transmission through the antenna subsystem;

e) the Antenna Controller Processor;

f) the Control and Monitoring Function.

maximum relative wind speed: addition of the magnitudes of the maximum wind speed and the maximum vehicle velocity

nominal antenna diameter: diameter of the antenna defined for the intended use of the VMES

NOTE 1: An antenna with a nominal antenna diameter *D* has the same technical characteristics as a circular aperture antenna of diameter *D*. For non circular aperture antenna, the nominal antenna diameter is defined for the intended use of the VMES.

NOTE 2: Network operators might request antennas of a certain diameter. Then an antenna that is compliant with the requirement for nominal antenna diameter equal to the requested antenna diameter can be used.

nominated bandwidth: bandwidth of the VMES radio frequency transmission nominated for the intended use of the VMES

NOTE 1: The nominated bandwidth is centred on the transmit frequency and does not exceed 5 times the occupied bandwidth.

NOTE 2: The nominated bandwidth is wide enough to encompass all spectral elements of the transmission which have a level greater than the specified spurious radiation limits. The nominated bandwidth is wide enough to take account of the transmit carrier frequency stability. This definition is chosen to allow flexibility regarding adjacent channel interference levels which will be taken into account by operational procedures depending on the exact transponder carrier assignment situation.

off-axis angle: angle between the direction of the axis of the antenna main beam and the considered direction

removable antenna: antenna which may be removed during the tests according to the EUT documentation

Response Channel (RC): channel by which VMES transmit monitoring information to the ACF

spurious radiation: any radiation outside the nominated bandwidth

transmission-disabled state: VMES is in this state when it is not authorized by the ACF to transmit

transmission-enabled state: VMES is in this state when it is authorized by the ACF to transmit

Wanted signal occupied Bandwidth (BW): for a digital modulation scheme: the width of the signal spectrum 10 dB below the maximum in-band power spectral density

3.2 Symbols

For the purposes of the present document, the following symbols apply:

dBc	ratio expressed in decibels relative to the e.i.r.p. of the unmodulated carrier
dBi	ratio of an antenna gain to the gain of an isotropic antenna, expressed in decibels
dBsd	ratio expressed in decibels relative to the spectral density
dBW	ratio of a power to 1 watt, expressed in decibels
dBpW	ratio of a power to 1 picowatt, expressed in decibels
dBµV/m	ratio of an electric field to 1 µV/m, expressed in decibels (20 log(electric field /1 µV/m))

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

ACF	Antenna Control Facility
BDC	Block Down Converter
Bo	occupied Bandwidth
BUC	Block Up Converter
BW	Bandwidth
CC	Control Channel
CCF	Control Channel reception Failure
CCR	Control Channel correctly Received
CENR	Cessation of Emissions Not Requested
CEPT	Conférence Européenne des Postes et Télécommunications (European Conference of Postal and Telecommunications Administrations)
CER	Cessation of Emissions Requested
CISPR	Comité International Spécial des Perturbations Radioélectriques (International Special Committee on Radio Interference)
CMF	Control and Monitoring Functions
CW	Continuous Wave
EIRP	Equivalent Isotropically Radiated Power
EME	Externally Mounted Equipment
EN	European Standard
EU	European Union
EUT	Equipment Under Test
FEC	Forward Error Correction
FSS	Fixed Satellite Service
g	gravity
grms	Root-Mean-Square Acceleration
GSO	Geostationary Satellite Orbit
HPA	High Power Amplifier
IME	Internally Mounted Equipment
LNB	Low Noise Block down converter
LO	Local Oscillator
NCF	Network Control Function
OATS	Open Area Test Site
OFDM	Orthogonal Frequency Division Multiplexing
RC	Response Channel
RF	Radio Frequency
SMF	System Monitoring Fail
SMP	System Monitoring Pass
SNR	Signal to Noise Ratio
STE	Special Test Equipment
TDMA	Time Division Multiple Access
TxD	Transmission Disable command
TxE	Transmission Enable command
VMES	Vehicle-Mounted Earth Station

4 Technical requirements specifications

4.1 General

4.1.1 Environmental profile

The technical requirements of the present document apply under the environmental profile for operation of the equipment, which shall be in accordance with its intended use, but as a minimum, shall be that specified in the test conditions contained in the present document. The equipment shall comply with all the technical requirements of the present document at all times when operating within the boundary limits of the operational environmental profile defined by its intended use which are given in Table 1.

Two classes of VMES are specified and the EUT documentation shall include designation of the class(es) under which they are compliant.

Table 1: Class A and B VMES Environmental Profiles

Parameter	Class A	Class B	
	Operational / Storage / Survival	Operational	Storage / Survival
High Temperature	+55 °C		
		+70 ºC	+85 ºC
Low Temperature	-25 °C	-55 ºC	
Temperature shock	3 °C/sec up to 40 °C	+85 ºC to -55 ºC	
Rain	[4]	- Blowing rain	
Humidity	25 °C to 55 °C, 100 % relative humidity	High temperature 60 ºC Low temperature 20 ºC 95 % relative humidity	
Salt Fog	[6]	2 cycles	
Sand and dust	[4]	Blowing dust and blowing sand	
Maximum Vehicle Velocity	To be provided with the EUT documentation; but in no case less than 80 km/h	To be provided with the EUT documentation; but in no case less than 80 km/h	
Acceleration		Structural only, 20 g	
Vibration	[5]	Composite wheeled vehicle vibration exposures Vertical 2,18 grms Lateral 1,6 grms Longitudinal 1,96 grms	
Shock	[7]		
		40 g during 11 ms	75 g during 6 ms
Icing / Freezing rain			13 mm build-up
Wind	100 km/h continuous, 160 km/h gusts	100 km/h continuous, 160 km/h gusts	
NOTE: Based on the above environmental requirements, annex C contains an indication of the typical scan rates and acceleration that would be necessary to achieve compliance. Figures defined for the Class B are further studied in MIL STD 810F [i.7].			

4.1.2 Operational configurations

Under operational conditions a VMES may dynamically change the occupied bandwidth and other transmission parameters (e.g. FEC, modulation, symbol rate) of the transmitted signal. For each occupied bandwidth an $EIRP_{max}$ and a nominated bandwidth shall be provided with the documentation of the EUT. For the purposes of verifying that the VMES complies with these specifications, the worst case combination of transmission parameters shall be identified. The following specifications apply to the VMES for each occupied bandwidth and other transmission parameters.

4.1.3 Determination of geographic location of the VMES

The VMES shall have a means of determining its geographic location within the accuracy defined for the intended use of the VMES and sufficient to maintain operation within all of the parameters defined for the intended use of the VMES.

4.1.4 VMES states and radio states

For the purpose of the present document the following four VMES states are defined, without presuming the effective implementation of the VMES state machine:

- "Non-valid";
- "Initial phase";
- "Transmission-disabled"; and
- "Transmission-enabled".

The four VMES states are represented on Figure 2 and are used in clause 4.2.8 for the specification of the Control and Monitoring Functions (CMFs).

In the "Non-valid" state and in the "Transmission-disabled" state the VMES is not allowed to transmit. In the "Transmission-enabled" state the VMES is allowed to transmit. In the "Initial phase" state the VMES is only allowed to transmit initial bursts or is waiting for a transmit enable/disable command.

The "Initial phase" is divided into three substates:

- "Initial phase-Standby" prior to the transmission of the first initial burst or when no initial bursts are transmitted;
- "Initial phase-BurstOn" during the transmission of the initial bursts;
- "Initial phase-BurstOff" between initial bursts.

NOTE: VMESs which do not transmit initial bursts have no "Initial phase-BurstOn" state and no "Initial phase-BurstOff" state.

The VMES "may transmit" when all the conditions for transmission are satisfied (e.g. in a state where transmissions are permitted, no failure detected, correctly pointed towards the satellite and there is no requirement for cessation of emissions).

The following radio states of the VMES are defined:

- "Emissions-disabled" when the VMES does not transmit any carrier;
- "Carrier-off" when the VMES may transmit and does not transmit any carrier;
- "Carrier-on" when the VMES may transmit and transmits a carrier.

Table 1a gives the only possible combinations of the VMES states and radio states which shall apply, with some examples of associated events.

Table 1a: VMES states and radio states of the VMES

VMES states and substates	Radio states	Examples of events
"Non-valid"	"Emissions-disabled"	After-power on; or After any failure; or During the checking phase.
"Initial phase"		When waiting for a transmission enable or disable command from the ACF.
"Initial phase-Standby"	"Emissions-disabled"	Before the first initial burst transmissions; or In locations where no transmission is allowed.
"Initial phase-BurstOn"	"Carrier-on"	During the transmission of each initial burst, and the pointing is correct.
"Initial phase-BurstOff"	"Carrier-off"	Between initial bursts; or When the pointing threshold is exceeded.
"Transmission-enabled"	"Carrier-off"	When no carrier is transmitted; or When receive synchronization is lost; or When the pointing threshold is exceeded.
	"Carrier-on"	During transmission of carrier(s), and the pointing is correct.
"Transmission-disabled"	"Emissions-disabled"	When a disable command from the ACF has been received and waiting for a transmission enable command from the ACF; or In locations where no transmission is allowed.

4.2 Conformance requirements

4.2.1 Off-axis spurious radiation

4.2.1.1 Justification

To limit the level of interference to terrestrial and satellite radio services in order to meet the essential requirements (Article 3.2 of the Radio Equipment Directive [i.6]) as referred to in the Scope of the present document.

4.2.1.2 Specification

The following specifications apply to the VMES transmitting at equivalent isotropically radiated power (e.i.r.p.) values up to and including $EIRP_{max}$.

1) The electric field strength level of any radiation from the VMES in the frequency range from 30 MHz to 1 GHz shall not exceed the limits specified in Table 1b.

Table 1b: Limits of radiated field strength at a test distance of 10 m in a 120 kHz bandwidth

Frequency range	Quasi-peak limits
30 MHz to 230 MHz	30 dBµV/m
230 MHz to 1 000 MHz	37 dBµV/m

The Quasi Peak detector shall be in accordance with CISPR 16-1-1 [2].

2) When the VMES is in the "Emissions-disabled" radio state, the off-axis spurious e.i.r.p. from the VMES, in any 100 kHz band, shall not exceed the limits in Table 2, for all off-axis angles greater than 7° or greater than the minimum elevation angle defined for the intended use of the VMES, whichever is lower.

Table 2: Limits of spurious e.i.r.p. - "Emissions-disabled" radio state

Frequency band	e.i.r.p. limit	Measurement bandwidth
1,0 GHz to 10,7 GHz	48 dBpW	100 kHz
10,7 GHz to 21,2 GHz	54 dBpW	100 kHz
21,2 GHz to 40,0 GHz	60 dBpW	100 kHz

The lower limits shall apply at the transition frequency.

3) The present document applies outside the nominated bandwidth. In the "Carrier-on" and "Carrier-off" radio states, the off-axis spurious e.i.r.p. density from the VMES, shall not exceed the limits in Table 3, for all off-axis angles greater than 7° or greater than the minimum elevation angle defined for the intended use of the VMES, whichever is lower.

Table 3: Limits of spurious e.i.r.p. - "Carrier-on" and "Carrier-off" radio states

Frequency band	e.i.r.p. limit	Measurement bandwidth	Remarks
1,0 GHz to 3,4 GHz	49 dBpW	100 kHz	
3,4 GHz to 10,7 GHz	55 dBpW	100 kHz	
10,7 GHz to 13,75 GHz	61 dBpW	100 kHz	
13,75 GHz to 14,0 GHz	95 dBpW	10 MHz	See note 1
14,00 GHz to 14,25 GHz			
14,25 GHz to 14,5 GHz			See note 2
14,5 GHz to 14,75 GHz	95 dBpW	10 MHz	See note 1
14,75 GHz to 21,2 GHz	61 dBpW	100 kHz	
21,2 GHz to 40,0 GHz	67 dBpW	100 kHz	

NOTE 1: This limit may be exceeded in a frequency band which shall not exceed 50 MHz, centred on the carrier frequency, provided that the on-axis e.i.r.p. density at the considered frequency is 50 dB below the maximum on-axis e.i.r.p. density of the signal (within the nominated bandwidth) expressed in dBW/100 kHz.
NOTE 2: In those CEPT countries where VMESs may not transmit in the band 14,25 GHz to 14,50 GHz, the e.i.r.p. limit defined for the band 14,5 GHz to 14,75 GHz shall apply.

The lower limits shall apply at the transition frequency.

In the frequency band 28,0 GHz to 29,0 GHz, for any 20 MHz band within which one or more spurious signals exceeding the above limit of 67 dBpW/100 kHz are present, then the power of each of those spurious signals exceeding the limit shall be added in watts, and the sum shall not exceed 78 dBpW.

4) These limits are applicable to the complete VMES equipment, comprising the IME and EME including cabling between the units.

4.2.1.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.2.

4.2.2 On-axis spurious radiation for VMES

4.2.2.1 Justification

To limit the level of interference to satellite radio services in order to meet the essential requirements (Article 3.2 of the Radio Equipment Directive [i.6]) as referred to in the Scope of the present document.

4.2.2.2 Specification

4.2.2.2.1 "Carrier-on" radio state

The following specification applies to the VMES transmitting at e.i.r.p. values up to $EIRP_{max}$.

The e.i.r.p. spectral density of the spurious radiation outside the nominated bandwidth and outside the 14,0 GHz to 14,5 GHz band shall not exceed 4 - K dBW in any 100 kHz band.

The e.i.r.p. spectral density of the spurious radiation outside the nominated bandwidth and within the range 14,0 GHz to 14,5 GHz, shall not exceed 18 - K dBW in any 100 kHz band.

Where K is the factor that accounts for a reduction on the on-axis spurious radiation level in case of multiple VMESs operating on the same frequency and the value of K is given by one of the following cases:

1) For the case where only one VMES transmits at any one time on a given carrier frequency, the value of K is 0.

2) For the case where several VMESs are expected to transmit simultaneously on a given carrier frequency at the same e.i.r.p. then K = 10 log (N) where N is the maximum number of these VMESs. The value of N shall not be exceeded for more than 0,01 % of the time. The value of N and the operational conditions of the system shall be defined for the intended use of the VMES.

3) For the case where several VMESs are expected to transmit simultaneously on a given carrier frequency at different e.i.r.p. levels then K = 10 log ($EIRP_{Aggregate}$/EIRP), where:

   - EIRP is the on-axis e.i.r.p. (Watts) of the VMES within the nominated bandwidth; and
   - $EIRP_{Aggregate}$ is the maximum on-axis aggregate e.i.r.p. (Watts) within the nominated bandwidth of the VMES system towards the satellite;
   - $EIRP_{Aggregate}$ shall not be exceeded for more than 0,01 % of the time.

The value of $EIRP_{Aggregate}$ and the operational conditions of the VMES network shall be defined for the intended use of the VMES.

For VMESs designed to transmit several carriers on different frequencies simultaneously (multicarrier operation), the above limits only apply to each individual carrier when transmitted alone.

4.2.2.2.2 "Carrier-off" and "Emissions-disabled" radio states

In the 14,0 GHz to 14,50 GHz band the e.i.r.p. spectral density of the spurious radiation outside the nominated bandwidth shall not exceed -21 dBW in any 100 kHz band.

4.2.2.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.3.

4.2.3 Off-axis e.i.r.p. emission density within the band

4.2.3.0 General

The present clause deals with the off-axis e.i.r.p. emission density (co-polarization and cross-polar) within the band 14,0 GHz to 14,50 GHz.

4.2.3.1 Justification

Protection of other satellite (uplink) systems in order to meet the essential requirements (Article 3.2 of the Radio Equipment Directive [i.6]) as referred to in the Scope of the present document.

4.2.3.2 Specification

The following specifications apply to the VMES transmitting at e.i.r.p. values up to $EIRP_{max}$.

The maximum e.i.r.p. in any 40 kHz band within the nominated bandwidth of the co-polarized component in any direction ϕ degrees from the antenna main beam axis shall not exceed the following limits:

$33 - 25 \log (\phi + \delta\phi) - H$ dBW for $2{,}5° \leq \phi + \delta\phi \leq 7{,}0°$;

$+12 - H$ dBW for $7{,}0° < \phi + \delta\phi \leq 9{,}2°$;

$36 - 25 \log (\phi + \delta\phi) - H$ dBW for $9{,}2° < \phi + \delta\phi \leq 48°$;

$- 6 - H$ dBW for $\phi + \delta\phi > 48°$;

where:

- ϕ is the angle, in degrees, between the main beam axis and the direction considered; and
- $\delta\phi$ is the pointing error threshold, in degrees, as defined for the intended use of the VMES (see clause 4.2.5.2).

For VMESs designed to transmit always at $EIRP_{max}$, H (in dB) is the maximum number of VMESs which may transmit at $EIRP_{max}$ in the same carrier frequency band as defined for the intended use of the VMES.

For VMESs designed to operate in an FSS network where the e.i.r.p. of each VMES is determined by the ACF and where the ACF is in charge of the compliance of the aggregate e.i.r.p. density with the above mask, H is the margin as defined for the intended use of the VMES for compliance with the mask when the VMES is transmitting at $EIRP_{max}$. In the case where the ACF uses the antenna pattern or the off-axis e.i.r.p. density the applicable pattern shall be defined for the intended use of the VMES, the value of H shall be set to 0 dB and the VMES e.i.r.p. density shall not exceed the e.i.r.p. density corresponding to the defined pattern. This margin H or this pattern may be a function of the position of the VMES relative to the GSO arc (e.g. in the case of active antennas).

For $\phi > 70°$ the values given above may be increased to 4 - H dBW over the range of angles for which the particular feed system may give rise to relatively high levels of spillover.

In addition the maximum e.i.r.p. in any 40 kHz band within the nominated bandwidth of the cross-polarized component in any direction ϕ degrees from the antenna main beam axis shall not exceed the following limits:

$23 - 25 \log (\phi + \delta\phi) - H$ dBW for $2{,}5° \leq \phi + \delta\phi \leq 7{,}0°$;

$2 - H$ dBW for $7{,}0° < \phi + \delta\phi \leq 9{,}2°$;

where ϕ and H are as defined above.

For non-continuous transmission, the above limits may not apply for a specific portion of each burst as defined for the intended use of the VMES. This excluded portion shall not exceed 50 µsec or 10 % of the burst, whichever is the smaller.

The excluded portion shall have characteristics similar to the remaining part of the burst:

- same symbol rate and modulation; and
- same or lower maximum amplitude.

NOTE 1: The specification above is intended to take account of the spectrum of the preamble of bursts in TDMA systems.

NOTE 2: The satellite operator may require lower off-axis e.i.r.p. limits in compliance with FSS intersystem coordination agreements. In all cases, off-axis e.i.r.p. emissions are subject to compliance with national regulations and the relevant FSS intersystem coordination agreements.

The antenna off-axis direction is defined by a pair of values (α, φ) where φ is the off-axis angle of that direction with the antenna main beam axis and α is the angle of the plane defined by that direction and the antenna main beam axis with any arbitrary plane containing the antenna main beam axis.

The range of values of φ and α is from φ_{min} - δφ to 180° for φ, and from -180° to +180° for α.

The above limits only apply to any off-axis direction (α, φ) within ± 3° of the visible part of the GSO. The concerned directions shall be any direction within the (α, φ) domain as defined for the intended use.

The determination of the (α, φ) subset shall take into account the operational conditions for which the VMES is designed, as defined for the intended use of the VMES or indicated within the user documentation. These conditions shall include:

- the range of latitudes of the VMES; if
- the minimum elevation pointing angle;
- the type of antenna mount (e.g. with azimuth and elevation axes or equatorial);
- the maximum static and dynamic alignment errors of the antenna mount axes;
- the maximum static and dynamic alignment errors of the antenna major axis with respect to the GSO arc;
- the range of adjustment for the major axis of the antenna for antennas with asymmetric main beam;
- the direction of the electric field radiated by the satellite with respect to the Earth's axis, when the electric field is used for the antenna alignment.

The alignment errors shall not exceed the declared maximum values when applying the alignment method defined for the intended use of the VMES.

NOTE 3: ETSI TR 102 375 [i.3] gives guidance for the determination of the concerned subset within the (α, φ) domain.

4.2.3.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.4.1 with the results being computed in accordance with clause 6.4.2.

4.2.4 Carrier suppression

4.2.4.1 Justification

To allow for the satisfactory suppression of transmissions of the VMES by the ACF, under any fault condition and under any cessation of emissions condition.

4.2.4.2 Specification

In the "Carrier-off" and in the "Emissions-disabled" radio states the on-axis e.i.r.p. density shall not exceed 4 dBW in any 100 kHz band within the nominated bandwidth.

4.2.4.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.5.

4.2.5 Antenna mechanical stability and pointing error

4.2.5.1 Justification

Protection of transmissions to adjacent satellites and cross-polarized transponders on the same satellite.

4.2.5.2 Specification

a) Mechanical stability:

- The VMES shall be designed to operate in accordance with the maximum values for the environmental parameters specified in clause 4.1.1.

b) Pointing error detection:

- The VMES shall be able to detect the pointing error. The VMES, when in the "Carrier-on" radio state, shall enter the "Carrier-off" radio state when the instantaneous pointing error exceeds the pointing error threshold, $\delta\phi$, relative to the direction of the wanted satellite at its actual position, within T seconds. The values of $\delta\phi$ and T shall be defined for the intended use of the VMES and the value of T shall not exceed 2 seconds. The VMES shall not re-enter the "Carrier-on" radio state until the instantaneous pointing error is within $\delta\phi$ for a period of $2 \times T$ seconds.

- The maximum time (IT_{max}) that the VMES can remain in "Transmission-enabled" state and "Carrier-off" radio state (see clause 4.2.8.8) shall be defined for the intended use of the VMES . If the pointing error threshold is exceeded for more than ITmax then the VMES shall enter the "Initial phase" state.

- The ranges of values for $\delta\phi$ as functions of on-axis e.i.r.p. spectral density such that the e.i.r.p. density limits in clause 4.2.3.2 are not exceeded shall be defined for the intended use of the VMES .

4.2.5.3 Conformance tests

Conformance tests are described in clauses 6.6 to 6.7. In addition, clause 6.11.8 contains the conformance test for inhibition of transmission when the outage time exceeds IT_{max}.

4.2.6 Cessation of emissions of the VMES

4.2.6.1 Justification

For cessation of emissions of the VMES where the VMES is not allowed to transmit.

4.2.6.2 Specification

4.2.6.2.1 Specification 1: Mode of cessation of emissions

At least one of the following two modes of cessation of emissions shall be implemented:

a) the ACF determines that the VMES shall cease emissions; and/or

b) the VMES autonomously determines that it shall cease emissions.

In addition, there shall be a "single-action" means (e.g. operating a switch) by which a local operator may disable the VMES and thereby cease emissions.

The VMES interfaces involved in the cessation of emissions, consistent with the intended use of the VMES, shall be provided with the documentation of the EUT:

- the list of relevant parameters which are collected by the VMES or the ACF for determination as to whether the VMES should cease emissions;
- the list of these relevant parameters which are used by the VMES;
- the list of these relevant parameters which are transmitted by the VMES to the ACF;
- the list of the relevant parameters which are received by the VMES from the ACF;
- for the collected relevant parameters, the VMES interface (s), including the protocols, the timing, the ranges of the values, the speed of the variations and the required accuracies;

- for the relevant parameters transmitted to the ACF, the VMES interface with the ACF, including the protocols and the timing;
- for the transmission parameter received from the ACF, the VMES interface with the ACF, including the protocols and the timing;
- these declared VMES interfaces shall be in accordance with the user documentation.

4.2.6.2.2 Specification 2: Conditions under which the VMES shall cease emissions

For VMES networks where the ACF determines completely or partially where and under which conditions the emissions of the VMES shall be ceased, the collection of the relevant parameters by the VMES and the exchange of information between the VMES and the ACF shall be sufficient for the ACF to determine when to cease emissions with the accuracy defined for the intended use of the VMES and to inform in time the VMES to cease emissions prior to entering any exclusionary zone.

For VMESs which determine completely where and under which conditions the emissions shall be ceased, the collection of the relevant parameters by the VMES shall be sufficient for the VMES to determine when to cease emissions within the accuracy defined for the intended use of the VMES.

The determination of these conditions shall take into account any inaccuracy in determination of the VMES geographic location and of the boundaries of the authorized operating areas in the data base used either by the VMES or the ACF, as defined for the intended use of the VMES.

The conditions for cessation of emissions shall take into account at least the following parameters:

- the location of the VMES and the boundaries of the authorized operating area;
- the operating parameters specified in clauses 4.2.1 through 4.2.6;
- the carrier frequency and the authorized frequency bands.

4.2.6.2.3 Specification 3: Cessation of emissions

A condition requiring cessation of emissions occurs either when the VMES receives the command from the ACF at its input or when the VMES determines autonomously on the need to cease emissions.

When in the "Transmission-enabled" state a condition requiring cessation of emissions occurs, the VMES shall cease transmissions and enter the "Initial Phase" state.

When in the "Transmission-disabled" state, the VMES shall not leave that state for the "Transmission-enabled" state as long as a condition requiring cessation of emissions exists or the last CC command received from the ACF is a transmission disable command.

When in the "Initial phase-BurstOn" or "Initial phase-BurstOff" substate a condition requiring cessation of emissions occurs, the VMES shall cease transmissions and enter the "Initial phase-Standby" substate.

When in the "Initial phase-Standby" substate, the VMES shall not leave that state for the "Initial phase-BurstOn" or the "Transmission-enabled" states as long as a condition requiring cessation of emissions exists.

The time for transition in any state from the occurrence of a condition requiring cessation of emissions to the "Emissions-disabled" radio state shall not exceed 1 second.

4.2.6.2.4 Specification 4: Fault conditions

Any collection of the relevant parameters by the VMES or transmission of these parameters to the ACF, which have not been completed correctly within the required delay(s) as defined for the intended use of the VMES, shall be considered as a fault condition.

Any transmission parameter not received or not correctly received from the ACF within the required delay defined for the intended use of the VMES shall be considered as a fault condition.

These fault conditions shall be processed as conditions requiring cessation of emissions.

4.2.6.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.9.

4.2.7 Identification of the VMES

4.2.7.1 Justification

To fulfil the requirements for identification of individual VMES units whether operated in a point-to-point fashion or in a network and for potential use by duly authorized entities.

4.2.7.2 Specification

The VMES shall be designed such that it is possible for the operator to identify which VMESs are transmitting with at least 100 m resolution in a given geographic area. The means by which identification of VMESs is accomplished shall be provided with the documentation of the EUT. In the case where an external system is required, any additional means necessary for identification of the VMES shall be provided with the documentation of the EUT.

4.2.7.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.10.

4.2.8 Control and Monitoring Functions (CMFs)

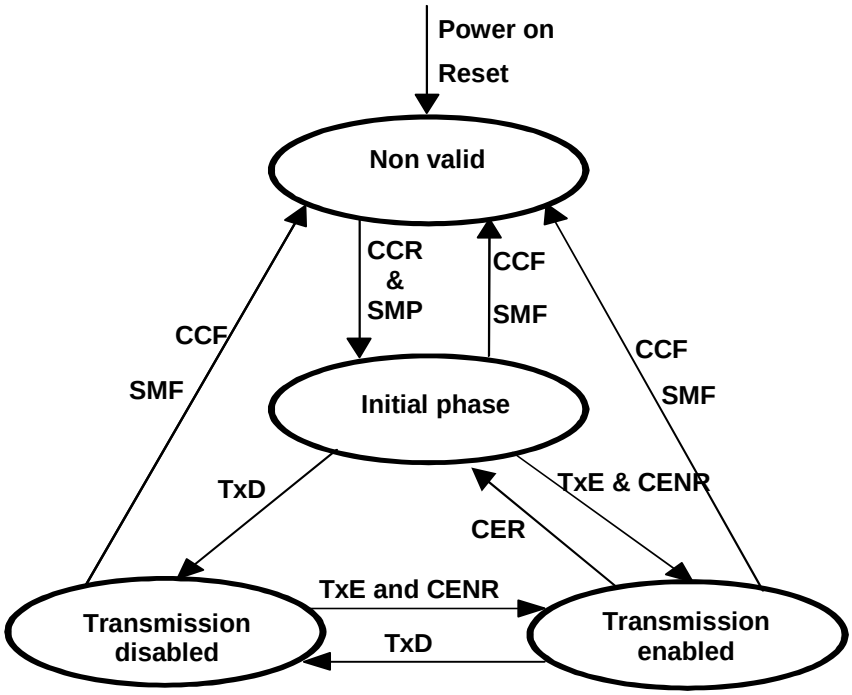
4.2.8.1 CMF state diagram

The following minimum set of CMFs shall be implemented in VMESs in order to minimize the probability that they originate unwanted transmissions that may give rise to harmful interference to other systems.

In the "Non-valid" state and in the "Transmission-disabled" state the VMES shall not transmit. In the "Transmission-enabled" state the VMES is allowed to transmit. In the "Initial phase" state the VMES is only allowed to transmit initial bursts.

NOTE 1: The restrictions in the "Initial phase" state are for the protection of other systems when the VMES is entering the system after a power-on or a reset. These initial burst restrictions do not apply to the VMES transmissions in the "Transmission-enabled" state and once a transmission enable command has been received by the VMES the VMES may transmit or not transmit as required.

Under any fault condition the VMES shall enter the "Non-valid" state where the e.i.r.p. limits for the "Emissions-disabled" radio state apply.



CER: Cessation of Emissions Required;
CENR: Cessation of Emissions Not Required;
SMP: System Monitoring Pass;
SMF: System Monitoring Fail;
TxE: Transmission Enable command;
TxD: Transmission Disable command;
CCR: Control Channel correctly Received;
CCF: Control Channel reception Failure.

Figure 2: State transition diagram of the control and monitoring function of a VMES

NOTE 2: From "Transmission-disabled" state a TxE command may also result in a transition towards the "Initial phase" state.

When the VMES transmits several carriers having different frequencies, a VMES state machine as described above may be associated with each carrier or each set of carriers. The events then apply to the subsystem associated with the specific carrier or the specific set of carriers, rather than the whole VMES.

4.2.8.2 Processor monitoring

4.2.8.2.1 Justification

To ensure that the VMES can suppress its transmissions in the event of a processor sub-system failure.

4.2.8.2.2 Specification

The VMES shall incorporate a processor monitoring function for each of its processors involved in maintaining the performance requirements of the present document and in Control and Monitoring Functions (CMF).

The processor monitoring function shall detect failures of these processors' hardware and software.

In the "Transmission-enabled" state, the VMES shall enter the "Non-valid" state or the "Carrier-off" radio state no later than 1 second after any fault condition occurs, and in any case it shall enter the "Non-valid" state within 30 seconds until the processor monitoring function has determined that all fault conditions have been cleared. In any other state after any fault condition occurs, the VMES shall enter the "Non-valid" state within 30 seconds until the processor monitoring function has determined that all fault conditions have been cleared.

4.2.8.2.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.11.2.

4.2.8.3 Transmit subsystem monitoring

4.2.8.3.1 Justification

To ensure the inhibition of transmissions that are potentially harmful to other systems in the event of incorrect operation of the transmit frequency generation sub-system.

4.2.8.3.2 Specification

The VMES shall monitor the operation of its transmit frequency generation sub-system and shall be able to detect:

a) loss of frequency lock;

b) absence of Local Oscillator (LO) output signal.

No later than 1 second after any of these fault conditions of the transmit frequency generation sub-system occurs, the VMES shall enter the "Non-valid" state until the transmit sub-system monitoring function has determined that all fault conditions have been cleared.

4.2.8.3.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.11.3.

4.2.8.4 Power-on/Reset

4.2.8.4.1 Justification

To demonstrate that the VMES achieves a controlled non-transmitting state following the powering of the unit, or the occurrence of a reset made by a local operator when this function is implemented.

4.2.8.4.2 Specification

Following a manual reset, when this function is implemented, the VMES shall enter the "Non-valid" state.

During and following "power-on" the VMES shall remain in the "Non-valid" state until all the conditions for entering "Initial phase" have been satisfied.

4.2.8.4.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.11.4.

4.2.8.5 Control Channel (CC) and Response Channel (RC)

4.2.8.5.1 Justification

Control Channels (CC) are used by VMESs to receive control information from the ACF and Response Channels (RCs) to respond to the ACF. For VMES designed to operate within networks where the ACF determines that the VMES shall cease emissions, then the CCs and RCs are also used for the dialogue with the ACF.

To ensure that the VMES cannot transmit unless it correctly receives the CC messages from the ACF.

4.2.8.5.2 Specification

4.2.8.5.2.1 Specification 1: types of CCs and RCs

a) The VMES shall have at least one CC from the ACF. If exchange of information with the ACF is necessary for operation of the VMES then the VMES shall have at least one RC to the ACF (see clause 4.2.6). Each CC and each RC may be internal or external.

The types (internal or external) of each CC and each RC shall be defined for the intended use of the VMES.

NOTE 1: The availability of the network carrying the external CC(s) and RC(s) and the numbers of external CC(s) and RC(s) are not within the scope of the present document.

NOTE 2: Some satellite operators may require that internal CC(s) and or RC(s) are available.

b) The connection between the ACF and the VMES via the CCs and RCs shall be either permanent or shall be set up on a call by call basis through a switched network. In case of connection through a switched network the VMES shall be able to receive calls from the ACF and to initiate calls towards the ACF in order to set up the CCs and RCs.

c) The VMES with an external CC shall not transmit without receiving an appropriate signal from the ACF through the satellite to which the VMES transmits, indicating to the VMES that the ACF is alive and ensuring that the VMES is pointing to the target satellite.

4.2.8.5.2.2 Specification 2: CC Reception

a) The VMES shall enter the "Non-valid" state immediately after a period not exceeding 30 seconds without correct reception of the CC from the ACF, i.e. without being able to receive CC messages from the ACF. This event is called a "CC disruption":

- In the case of external CCs and RCs without permanent connection of the ACF with the VMES, the ability to receive CC messages from the ACF is the ability at any time to receive calls and messages within the timing requirements of the present document from the network through which is connected the ACF.

- The inability to receive CC messages from the ACF may be due to the following various causes but not limited to them: no received signal from the ACF or from the network, a too low level received signal, no network accessible, the inability to lock onto the received carrier frequency, to demodulate, to decode, to receive calls and/or messages, a hardware failure or power off.

b) The VMES shall remain in the "Non-valid" state as long as the VMES is unable to receive CC messages from the ACF.

c) From the "Non-valid" state the VMES may enter the "Initial phase" state if the following conditions are met:

- the VMES is able to receive CC messages from the ACF; and

- no fault conditions are present.

4.2.8.5.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.11.5.

4.2.8.6 Network control commands

4.2.8.6.1 Justification

These requirements ensure that the VMES is capable of:

a) retaining a unique identification in the network;

b) receiving commands from the ACF through its CC(s) and executing those commands.

4.2.8.6.2 Specification

The VMES shall hold, in non-volatile memory, its unique identification code in the network.

The VMES shall be capable of receiving through its CCs dedicated messages (addressed to the VMES) from the ACF, and which contain:

- Transmission Enable commands (TxE);
- Transmission Disable commands (TxD).

When in the "Initial phase" or "Transmission-enabled" states, once a transmission disable command is received, within 1 second the VMES shall enter into, and shall remain in the "Transmission-disabled" state until the transmission disable state is superseded by a subsequent transmission enable command (see also clause 4.2.6).

When in the "Initial phase" or "Transmission-disabled" states, once a transmission enable command is received, the VMES may enter into the "Transmission-enabled" state.

When entering the "Initial phase" from the "Non-valid" state, the last TxE or TxD command received from the ACF may be used by the VMES to enter the "Transmission-enabled" state or the "Transmission-disabled" state, respectively, if since the time of reception of that command no "CC disruption" occurred.

NOTE: The physical unit in charge of the reception of the CCs, of the transmissions of the RCs and of the CC commands reception may be a separate and independent unit from the other units of the VMES and it may be common to several VMESs.

4.2.8.6.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.11.6.

4.2.8.7 Initial burst transmission

4.2.8.7.1 Justification

Restrictions on the initial burst transmissions are necessary to limit disturbance to other services.

4.2.8.7.2 Specification

For systems where no transmission enable command is foreseen without request from the VMES, in the "Initial phase" state the VMES may transmit initial bursts:

a) The VMES shall not start the transmission of initial bursts before having obtained a confirmation that no cessation of emission applies where the VMES is located. This confirmation shall be obtained either by local means of determination of the cessation of emissions (see clause 4.2.6) or from the ACF via an external control channel.

b) The duty cycle of the burst retransmission shall not exceed 0,2 %, where the duty cycle is defined as the ratio of burst duration to the duration between two successive bursts.

c) Each burst shall not carry more than the minimum number of data bytes excluding the burst preambles and the FEC coding bits necessary to establish synchronization.

d) The initial burst shall be transmitted at an e.i.r.p. no greater than $EIRP_{max}$.

e) The duration between two successive bursts shall not be less than the required ACF response time as defined for the intended use of the VMES. This response time is defined as the duration, measured at the VMES, between an initial burst transmission and the reception and processing of a transmission enable or disable command from the ACF which is never exceeded during 99 % of the cases under normal conditions in the system for which the VMES is designed, as defined for the intended use of the VMES.

4.2.8.7.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.11.7.

4.2.8.8 Inhibition of transmissions

4.2.8.8.1 Justification

To ensure the correct inhibition of transmissions that are potentially harmful to other systems and persons in the event of signal blockage.

4.2.8.8.2 Specification

The VMES shall inhibit transmissions whenever there is a loss of receive carrier (e.g. synchronization or signal level).

VMES shall enter the "Carrier-off" radio state within a period not exceeding 0,5 seconds, when these conditions persist during that period, and shall remain in this radio state until the receiver carrier has been restored.

The maximum outage time (IT_{max}) for this condition ("Transmission-enabled" state and "Carrier-off" radio state) shall be provided with the documentation of the EUT. If the receive carrier is not restored within IT_{max}, then the VMES shall transition to the "Initial phase" state.

The time to transition from "Carrier-off" to "Carrier-on" radio state (IT_{trans}) when in "Transmission-enabled" state after the receive carrier is restored within IT_{max}, shall be provided with the documentation of the EUT.

4.2.8.8.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.11.8.

4.2.9 Receive antenna off-axis gain pattern

4.2.9.1 Justification

To protect the wanted signals from interference from terrestrial services and from other satellite services in order to meet the essential requirements (Article 3.2 of the Radio Equipment Directive [i.6]) as referred to in the Scope of the present document.

4.2.9.2 Specification

The maximum antenna gain of each of the co-polarized components in any direction φ degrees from the antenna main beam axis shall not exceed the following limits:

$G = 32 - 25 \log \varphi$ dBi for $\varphi_{min} \leq \varphi < 48°$

$G = -10$ dBi for $48° \leq \varphi \leq 85°$

$G = 0$ dBi for $85° \leq \varphi \leq 180°$

where:

φ_{min} = 1° or 100 λ/*D* degrees, whichever is the greater, for $D/\lambda \geq 50$.

φ_{min} = 2° or 114 $(D/\lambda)^{-1,09}$ degrees, whichever is the greater, for $D/\lambda < 50$.

D is the nominal diameter of the antenna.

In addition the maximum antenna gain of each of the cross-polarized components in any direction ϕ degrees from the antenna main beam axis shall not exceed the following limits:

$$G_x(\varphi) = 23 - 20 \log \varphi \text{ dBi for } \varphi_r \leq \varphi \leq 7°$$

where φ_r is equal to 1° or 100 λ/*D*, whichever is greater.

4.2.9.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.12.

4.2.10 Blocking performance

4.2.10.1 Justification

To prevent high power signals outside the receive frequency band from blocking the reception of signals inside the receive frequency band.

4.2.10.2 Specification

Receiver blocking is characterized here through gain compression for a signal inside the receive frequency band that is caused by another signal outside the receive frequency band at high power. The level of the other signal is compared to the level of a signal inside the receive frequency band that would cause the same gain compression.

Receiver blocking rejection at a particular frequency is defined as the level of a second signal at this frequency that causes a certain gain compression to a first signal inside the receive frequency band, minus the level of a second signal at a frequency inside the receive frequency band that causes the same gain compression.

The first signal shall be at the centre frequency of the receive frequency band and have the same level as the wanted signal, as defined for its intended use. The second signal shall cause a gain compression of 1 dB for the first signal.

The rejection shall comply with Table 4.

Table 4: Receiver blocking rejection

Frequency	Minimum rejection
below 9 GHz	20 dB
9 to 10 GHz	10 dB
14 to 16 GHz	10 dB
Above 16 GHz	20 dB

4.2.10.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.13.

4.2.11 Adjacent Signal Selectivity

4.2.11.1 Justification

To enable reception of a wanted signal in presence of other signals on adjacent frequencies which are transmitted with high EIRP density from target satellite orbital position.

NOTE: The power level of signals transmitted from the same orbital position are under control of the satellite operator. Signals transmitted from an adjacent orbital position that is not near-by are suppressed by the antenna gain pattern.

4.2.11.2 Specification

Adjacent Signal Selectivity is a measure of a receiver's ability to receive a signal at its assigned channel frequency in the presence of an adjacent signal at a given frequency offset from the centre frequency of the assigned channel.
The adjacent signal shall occupy the same bandwidth as the wanted signal. Frequency offset and relative power level of the adjacent signal compared to the wanted signal shall take the value given in Table 5. BW is the wanted signal occupied bandwidth.

Table 5: Adjacent signal frequency and Power level

Signal	Centre frequency offset from wanted signal	Power level relative to wanted signal
Adjacent Carrier	BW	7 dBsd

The decrease in the required signal to noise ratio in the presence of an adjacent signal shall be no more than 0,5 dB.

4.2.11.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.14.

5 Testing for compliance with technical requirements

5.1 Environmental conditions for testing

Survivability testing shall be carried out prior to tests on the radiation and stabilization subsystems.

Tests defined in the present document shall be carried out at representative points within the boundary limits of the declared operational environmental profile, defined by its intended use, which, as a minimum, shall be that specified in the test conditions contained in the present document.

Where technical performance varies subject to environmental conditions, tests shall be carried out under a sufficient variety of environmental conditions as specified in the present document to give confidence of compliance for the affected technical requirements.

5.2 Essential radio test suites

The essential radio test suites for a complete VMES are given in clause 6.

6 Test methods for all aspects of the VMES

6.1 General

The present clause describes the general requirements for verifying that the performance of the EUT complies with the specifications. The documentary evidence of the results obtained in performing the essential radio tests shall be maintained with the EUT documentation. At a minimum, the documentary evidence shall include:

- test setup (configurations, test equipment and calibration status);
- test conditions (environmental and operational parameters);
- method of testing;
- results of measurements, measurement resolution and uncertainty; and
- statement of compliance with technical specifications of the present document.

The test conditions for each test shall be representative of the intended operational environment and be noted in the test report where it has a significant effect on the measurement results.

The type, termination and length of each cable used shall be representative of the intended installation and recorded in the test report. For maximum cable lengths longer than 10 m, as defined for the intended use of the VMES, the tests shall be performed with cables no shorter than 10 m.

ETSI TR 102 215 [i.4] provides a recommended approach to measurement uncertainty and interpretation of results. Where the measurement approach does not conform to the guidance in ETSI TR 102 215 [i.4], the EUT shall be provided with documentation to justify the uncertainty achieved under this method.

The test site and the method of measurement shall also be in accordance with CISPR 16-1-5 [3] as applicable.

Where a radome is required in operation, the tests shall be conducted with the radome in place or, if testing with the radome in place is impractical, then an appropriate analytic technique shall be used to compensate for the absence of the radome.

To enable the performance tests to be carried out the use of an ACF or a Special Test Equipment (STE), made available with the EUT, may be necessary. Since this STE will be specific for the particular system, it is not possible to provide detailed specifications in the present document. However, the following baseline is provided:

- if the VMES requires to receive a modulated carrier from the satellite in order to transmit, then special test arrangements are required to simulate the satellite signal, thus enabling the VMES to transmit allowing measurement of transmission parameters;
- any characteristic of these special test arrangements which may have direct or indirect effects on the parameters to be measured shall be defined for the intended use of the VMES.

All tests with carrier-on shall be undertaken with the transmitter operating at $EIRP_{max}$, as per the specific requirement, and with the normal radio operating parameters, as defined for the intended use of the VMES.

If the Equipment Under Test (EUT) is a VMES that requires hardware and/or software modification(s)for these tests then full documentation of such modification(s) shall be provided to demonstrate that the modification(s) will simulate the required test condition, without its main characteristics being changed.

6.2 Off-axis spurious radiation

6.2.0 General

The tests for the VMES specification in clause 4.2.1.2 shall be conducted in "Carrier-on", "Carrier-off" and "Emissions-disabled" radio states as required. The tests in "Carrier-on" radio state shall be undertaken with the transmitter operating at $EIRP_{max}$.

6.2.1 Test method

The EUT documentation shall include any means to identify frequencies of off-axis spurious radiation and to measure (or calculate) the radiated power levels of identified spurious radiations.

NOTE: For the purposes of this procedure, the measuring antenna is aligned in the polarization which produces the largest response between the EUT and the measuring antenna.

For frequencies up to 80 MHz the measuring antenna shall be a balanced dipole with a length equal to the 80 MHz resonant length and shall be matched to the feeder by a suitable transforming device. Measurements with broad band antennas are also possible provided that the test site has been calibrated according to the requirements of CISPR 16-1-5 [3].

For frequencies between 80 MHz and 1 000 MHz the measuring antenna shall be a balanced dipole which shall be resonant in length. Measurements with broad band antennas are also possible provided that the test site has been calibrated according to the requirements of CISPR 16-1-5 [3].

For frequencies above 1 000 MHz the antenna shall be a horn radiator of known gain/frequency characteristics. When used for reception the antenna and any associated amplification system, the combined amplitude/frequency response calibration curves shall be verified across the measurement frequency range considered for the antenna, to account for the interaction between the components. The antenna is mounted on a support capable of allowing the antenna to be used in either horizontal or vertical polarization and at the specified height.

Unused input and output ports of the EUT shall be connected to cables (see clause 6.1) and terminated to simulate the impedance of the relevant ports of the radio equipment. These cables shall be bundled at the approximate centre of the cable with the bundles of 30 cm to 40 cm in length running in the horizontal plane from the port to which it is connected. If it is impractical to do so because of cable bulk or stiffness, the disposition of the excess cable shall be precisely noted in the test report.

The radiations shall be measured in the frequency range of 30 MHz to 2 GHz in accordance with CISPR 16-1-5 [3] using the measuring receiver or a comparable spectrum analyser. During the measurements up to 1 GHz the quasi peak detector shall be used, for measurements in the frequency range 1 GHz to 2 GHz a peak detector shall be used.

6.3 On-axis spurious radiation

6.3.0 General

The tests for the VMES specification in clause 4.2.2.2 shall be conducted in "Carrier-on", "Carrier-off" and "Emissions-disabled" radio states as required.

The tests in "Carrier-on" radio state shall be undertaken with the transmitter operating at $EIRP_{max}$. Test measurements shall be made at least three points within the operating band including: locating the carrier at the centre frequency of the operating band; as close to the lower operating frequency limit as possible; and as close to the upper operating frequency limit as possible.

6.3.1 Test method

6.3.1.1 General

The tests shall be undertaken with the transmitter operating at $EIRP_{max}$.

NOTE: For the purposes of this procedure, the measuring antenna is aligned in the polarization which produces the largest response between the EUT and the measuring antenna.

For EUT equipment for which measurements at the antenna flange are not possible, the measurements shall be performed with a test antenna.

For EUT equipment for which measurements at the antenna flange are possible, the measurements shall be performed at the antenna flange. The EUT is the VMES with its antenna comprising both the internal and external units interconnected by appropriate cables.

6.3.1.2 Method of measurement at the antenna flange

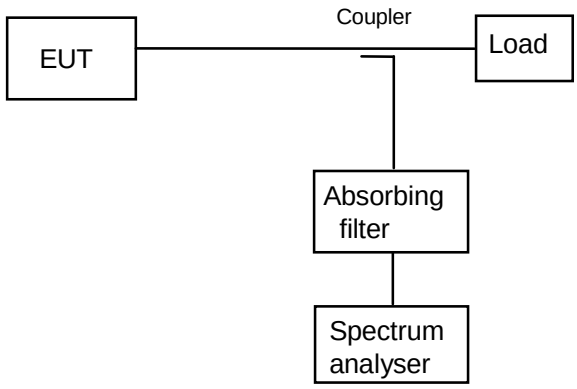


Figure 3: Test set up - on-axis spurious radiation measurements at the antenna flange

a) The test set up shall be as shown in Figure 3. In order to protect the spectrum analyser an absorbing filter can be used and it shall be tuned to the transmit carrier frequency.

b) The EUT shall transmit one modulated carrier continuously at its maximum burst rate, centred on a frequency within the lower limit of the operating frequency band of the EUT. The EUT shall be operated at the highest normal operating e.i.r.p. The frequency range 14,00 GHz to 14,5 GHz shall be measured.

c) Due to the proximity of the carrier the spectrum analyser resolution bandwidth shall be set to a measurement bandwidth of 3 kHz, or lower. If the measurement bandwidth is different from the specified measurement bandwidth, linear bandwidth correction shall be performed for noise-like wideband spurious radiation.

d) To obtain the on-axis spurious e.i.r.p., the antenna transmit gain, as defined for its intended use, shall be added to any figure obtained in the above measurement and any correction or calibration factor summated with the result.

e) The tests in b) to e) shall be repeated with a transmit frequency in the centre of the operating frequency band.

f) The tests in b) to e) shall be repeated with a transmit frequency close to the upper limit of the operating frequency band of the EUT.

g) The test shall be repeated in the "Carrier-off" radio state.

h) The test shall be repeated in the "Emissions disabled" radio state.

6.4 Off-axis e.i.r.p. emission density within the band

6.4.0 General

The tests for the VMES specification in clause 4.2.3.2 for off-axis e.i.r.p. emission density (co-polarization and cross-polarization) within the band 14,0 GHz to 14,50 GHz shall be conducted in the "Carrier-on" radio state. The tests shall be undertaken with the transmitter operating at $EIRP_{max}$.

The elevation angle within the range of applicable angles, with respect to the local horizon, at which the worst case radiation pattern performance and losses occur shall be provided with the documentation of the EUT, and tests shall be carried out at this elevation angle.

The tests in "Carrier-on" radio state shall be undertaken with the transmitter operating at $EIRP_{max}$. Test measurements shall be made at least three points within the operating band including: locating the carrier at the centre frequency of the operating band; as close to the lower operating frequency limit as possible; and as close to the upper operating frequency limit as possible.

6.4.1 Test method

6.4.1.1 General

The EUT documentation shall include any means to identify frequencies of off-axis radiation and to measure (or calculate) the radiated emission power density levels within the band.

NOTE: For the purposes of this procedure, the measuring antenna is aligned in the polarization which produces the largest response between the EUT and the measuring antenna.

The measurement shall be made in the operational configuration(s) (including occupied bandwidth, $EIRP_{max}$) producing the highest emission density in the frequency band. The selected configuration(s) shall be recorded in the test report.

If the test is to be carried out with an STE, then the STE shall provide all the signals needed by the VMESs under normal operation (e.g. a radio beacon if it is expected to be received by the satellite).

The e.i.r.p. densities are determined from the measurements of the antenna copolarization and crosspolarization gain patterns, and of the power density at the antenna flange. These e.i.r.p. densities shall be compared to the specified masks of clause 4.2.3.2.

To ascertain the off-axis e.i.r.p. it is necessary to know the transmit power density and antenna transmit radiation pattern. To ascertain the radiation pattern it is necessary to know the antenna transmit gain.

The following three measurement procedures shall, therefore, be performed:

a) transmit output power density (dBW/40 kHz);

b) antenna transmit gain (dBi);

c) antenna transmit radiation patterns (dBi).

6.4.1.2 Transmit output power density

The EUT output power shall be set to the power corresponding to $EIRP_{max}$. The measurements shall be made in the operational configuration(s) producing the highest emission density in the frequency band. The selected configuration(s) shall be recorded in the test report.

For purpose of the test, the EUT is the VMES fitted with its antenna.

The distance between the EUT or the substitution antenna and the measuring antenna shall be such that the radiating near-field of each antenna shall not overlap with that of the other. The larger radiating near-field of the EUT and substitution antenna shall be used to determine the minimum distance between the EUT and measuring antenna in the first instance.

6.4.1.2.1 Test site

This test shall be performed on either an outdoor far-field test site or compact test range. The preferred test site is the compact test range.

6.4.1.2.2 Method of measurement

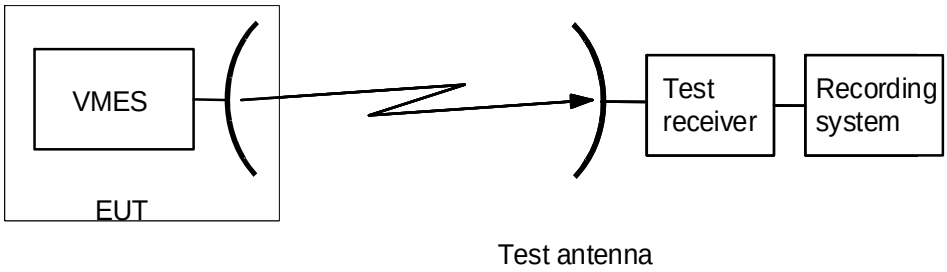


Figure 4: Test set up - maximum on-axis EIRP received measurement

a) The test set up shall be as shown in Figure 4, both antennas being in line of sight of each other. A signal proportional to the angular position from the servomechanism and the signal level from the test receiver proportional to the power received in dBW shall be applied to the recording system.

b) The test shall be carried out with the EUT transmitting the operational waveform signal, at maximum power, which carries a pseudorandom data stream. The transmission shall be continuous where possible.

c) The resolution bandwidth of the spectrum analyser shall be set larger but as close as possible to the Bo of the transmitted signal.

d) The ground plane axis shall be pointed horizontal towards the STE. The beam Zn-angle and the beam Az-angle shall be set to 0°.

e) The initial E-plane of the test signal radiated by the EUT through its antenna shall be horizontal. The test receiver antenna main beam axis shall be aligned with the main beam axis of the EUT antenna. The test receiver antenna and its polarizer shall be rotated around its main beam axis such that its E-plane coincides with the E-plane of the EUT.

f) After any change of the E-plane of the test signal, the test receiver antenna shall be rotated around its main beam axis such that its E-plane coincides with the E-plane of the EUT.

g) The frequency of the test signal shall be set 0,5 × Bo above the bottom of the lowest frequency band specified for the intended use of the VMES.

h) The EUT shall be aligned to maximize the received signal and the recording system adjusted to give the maximum reading on the chart.

i) The EUT shall be driven in azimuth in one direction through 10°.

j) The pattern measurement is then obtained by driving the EUT in azimuth back through boresight to 10° the other side with the plotter recording the results.

k) The EUT shall be replaced by the signal generator connected to a calibrated antenna (a substitution antenna), transmitting a carrier at a frequency equal to the EUT carrier frequency. The received signal level shall be maximized.

l) This level shall be recorded on the recording system.

m) The signal generator with the substitution antenna shall be driven in azimuth as in h) and j).

n) The EIRP of the signal radiated by the EUT shall be calculated from:

$$EIRP_{EUT} = L_1 - L_2 + G + P$$

where:

- $EIRP_{EUT}$ is the EIRP of the signal radiated by the EUT (dBW), in the considered direction;
- L_1 is the level obtained with the EUT (dBW), in the considered same direction;
- L_2 is the level obtained with the signal generator with the substitution antenna (dBW);
- G is the calibrated gain of the substitution antenna at the test frequency (dBi);
- P is the power produced by the signal generator at the flange of the substitution antenna (dBW).

o) The tests in h) to n) shall be repeated with the frequency changed to the middle of the lowest frequency band specified for the intended use of the VMES.

p) The tests in h) to n) shall be repeated with the frequency changed to 0,5 × Bo below the top of the lowest frequency band specified for the intended use of the VMES.

q) The tests in h) to p) may be performed simultaneously.

r) The tests in h) to q) shall be repeated with the E-plane vertical.

s) The tests in h) to q) shall be repeated with the E-plane at +45° to the horizontal plane.

t) The tests in h) to q) shall be repeated with the E-plane at -45° to the horizontal plane.

u) For an active antenna the tests in h) to t) shall be repeated with the beam Zn angle increased in 15° increments up to the maximum beam Zn angle specified for the intended use of the VMES when maximum EIRP variation is expected with the beam Zn angle.

v) For an active antenna the tests in h) to u) shall be repeated with the beam Az angle increased in 45° increments when maximum EIRP variation is expected with the beam Az angle.

w) The tests in g) to v) shall be repeated for all frequency bands specified for the intended use of the VMES.

x) The maximum on-axis EIRP of the signal radiated by the EUT is the maximum value of the values calculated in step n).

6.4.1.3 Antenna transmit gain

For the purpose of the present document, the antenna transmit gain is defined as the ratio, expressed in decibels (dBi), of the power that would have to be supplied to an isotropic radiator isolated in space, to the power supplied to the antenna being considered, so that they produce the same field strength at the same distance in the same direction. Unless otherwise specified the transmit gain is for the direction of maximum radiation (i.e. peak transmit gain).

This test is performed with the antenna removed and the measurements made at the antenna flange. For an EUT where the antenna is integrated within the unit, the EUT documentation shall include any means to derive the antenna transmit gain.

6.4.1.4 Antenna transmit radiation patterns

6.4.1.4.0 Test site and Test arrangement

For the purpose of the present document, the antenna transmit radiation patterns are diagrams relating field strength to the angle of the direction pointed by the antenna at a constant large distance from the antenna (in the far field of the antenna).

Near field measurements can also be used if the appropriate correction factors can be applied. The EUT documentation shall include evidence and any means for performing these measurements and justification for adopting this procedure for these particular measurements.

Tests shall be performed and the results recorded for co- and cross-polarization antenna transmit radiation patterns and in the elevation and azimuth planes, with the antenna radiating through its radome at the minimum elevation angle as defined for the intended use of the VMES. The co-polarization transmit pattern measurement shall be performed in azimuth from -180° to +180° with respect to the antenna mechanical boresight and from -15° to +90° in elevation with respect to the antenna mechanical boresight, with the results recorded. The cross-polarization transmit pattern measurement shall be performed in azimuth from -10° to +10° and from -1° to +10° in elevation with the results recorded. In the case of antennas where the sidelobes vary as a function of the elevation angle, the elevation transmit radiation pattern measurements shall be made with the antenna set to the elevation angle that produces the maximum sidelobe level.

This test shall be performed using the method of measurement as specified in clause 6.4.1.4.1.

The tests can be carried out in an Open Area Test Site (OATS) environment or in an anechoic chamber or in a semi-anechoic chamber as they lead to comparable test results for establishing compliance. Ambient noise levels shall be at least 6 dB below the unwanted emissions limit. The preferred test site is the semi-anechoic chamber.

6.4.1.4.1 Method of measurement

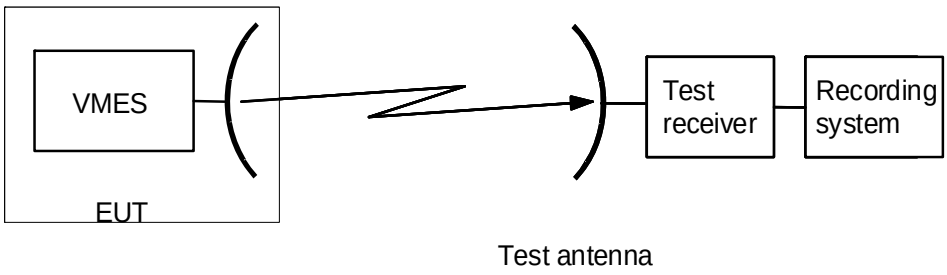


Figure 5: Test set up - antenna transmit radiation pattern measurement

a) The test arrangement shall be as shown in Figure 5, both antennas being in line of sight of each other. A signal proportional to the angular position from the servomechanism and the signal level from the test receiver proportional to the power received in dBW shall be applied to the recording system.

b) The test shall be carried out with the EUT transmitting the operational waveform signal, which carries a pseudorandom data stream. The transmission shall be continuous where possible.

c) The resolution bandwidth of the spectrum analyser shall be set with the Bo of the transmitted signal.

d) The antenna main beam axis shall be pointed horizontal towards the STE. The beam Zn-angle and the beam Az-angle shall be set to 0°.

e) The initial E-plane of the test signal radiated by the EUT through its antenna shall be horizontal. The test receiver antenna main beam axis shall be aligned with the main beam axis of the EUT. The test receiver antenna and its polarizer shall be rotated around its main beam axis such that its E-plane coincides with the E-plane of the EUT.

f) After any change of the E-plane of the EUT test signal, the test receiver antenna shall be rotated around its main beam axis such that its E-plane coincides with the E-plane of the EUT.

g) The frequency of the EUT test signal shall be set 0,5 × Bo above the bottom of the lowest frequency band specified for the intended use of the VMES.

h) The EUT shall be aligned to maximize the received signal and the recording system adjusted to give the maximum reading on the chart.

i) The EUT shall be driven in azimuth to -180°.

j) The transmit pattern measurement is then obtained by driving the EUT in azimuth through from -180° to +180° with the plotter recording the results. Measurements can be excluded for the range of angles where the radiation pattern from the antenna is fully blocked by the vechicle for the models specified for the intended use of the VMES. Exclusions shall be noted in the test report.

k) The tests in h) to j) shall be repeated with the frequency changed to the middle of the lowest frequency band specified for the intended use of the VMES.

l) The tests in h) to j) shall be repeated with the frequency changed to 0,5 × Bo below the top of the lowest frequency band specified for the intended use of the VMES.

m) The tests in h) to l) may be performed simultaneously.

n) The tests in f) to m) shall be repeated with the E-plane vertical.

o) The tests in f) to m) shall be repeated with the E-plane at +α to the horizontal plane. α is defined as the angle (°) between the horizontal plane and the geostationary orbital arc, as seen from the range of latitudes as specified for the intended use of the VMES.

p) The tests in f) to m) shall be repeated with the E-plane at -α to the horizontal plane. α is as defined in k).

q) For an active antenna, the tests in f) to p) shall be repeated with beam Zn angle increased in 15° increments up to the maximum beam Zn angle specified for the intended use of the VMES when maximum EIRP variation is expected with beam Zn angle.

r) For an active antenna, the tests in f) to q) shall be repeated with the beam Az angle increased in 45° increments when maximum EIRP variation is expected with the beam Az angle.

s) The tests in f) to r) shall be repeated for all frequency bands specified for the intended use of the VMES.

6.4.2 Computation of results

The results shall be computed by producing a "mask" to the specified limits with the reference level being equal to the sum of the transmitter output power density and the gain of the antenna. This reference shall then be placed on the maximum point of the plots obtained from the transmit radiation pattern measurements, so as to ascertain that the off-axis e.i.r.p. density is within the mask and thus conforming to the specification in clause 4.2.3.

6.5 Carrier suppression

6.5.1 Test method

For the purposes of this test, the EUT shall transmit one carrier modulated continuously, or at its maximum burst rate where applicable, centred on the middle frequency of the operating frequency band as defined for the intended use of the VMES. The "transmission-disabled state" shall be obtained by use of the ACF or an STE, if required. For conducted measurements the maximum residual carrier power density within the nominated bandwidth shall be measured and added to the antenna on-axis gain. For radiated measurements the maximum residual e.i.r.p. density within the nominated bandwidth shall be measured and recorded. If the results obtained are within the limit given in clause 4.2.4.2, then the EUT shall be declared compliant with the requirements.

The test shall follow the procedure below:

a) The test set up for radiated measurements shall be as shown in Figure 5.

b) The EUT shall transmit one modulated carrier continuously, or at its maximum burst rate, centred on the middle frequency of the operating frequency band as defined for the intended use of the VMES.

c) The resolution bandwidth of the spectrum analyser shall be set to 3 kHz.

d) The "transmission disabled state" shall be obtained by use of the NCF.

e) For conducted measurements the maximum residual carrier power density within the nominated bandwidth shall be measured and added to the antenna on-axis gain.

f) For radiated measurements the maximum residual e.i.r.p. density within the nominated bandwidth shall be measured and recorded.

An STE provided with the EUT can be used to enable the VMES transmission to be suppressed.

6.6 Antenna pointing for VMES

6.6.1 General

For the purpose of these tests the EUT is the VMES with its antenna connected.

6.6.2 Test method

The means for inducing pointing errors shall be representative of actual operating vehicle dynamics, within the dynamic limits defined for the intended use of the VMES. Pointing errors shall be induced in at least two orthogonal directions; but not necessarily during the same test sequence. The means of inducing pointing errors shall be included in the test report.

After the pointing error has been decreased below the pointing error threshold, it shall be verified that the EUT does not re-enter the "Carrier-on" radio state for at least 2 × T seconds, as specified in clause 4.2.5.

The tests shall be repeated 5 times to ensure reliability of pointing error detection and appropriate response.

The antenna pointing test method is designed to verify the ability of the EUT to detect and correctly respond to pointing errors that exceed the threshold, $\delta\phi$, defined for the intended use of the VMES as required in clause 4.2.5. For the purposes of testing antenna pointing, the documentation provided with the EUT shall include:

- the operational dynamic limits for ship motion under which the VMES can maintain tracking within the defined $\delta\phi$;
- the survivability dynamic limits for operation without permanent damage; and
- the dynamic limits for testing of the EUT, which shall not be less than the operational dynamic limits and not greater than the survivability dynamic limits.

For the purpose of these tests the EUT is the VMES with its antenna connected. The test set up is provided in Figure 6.

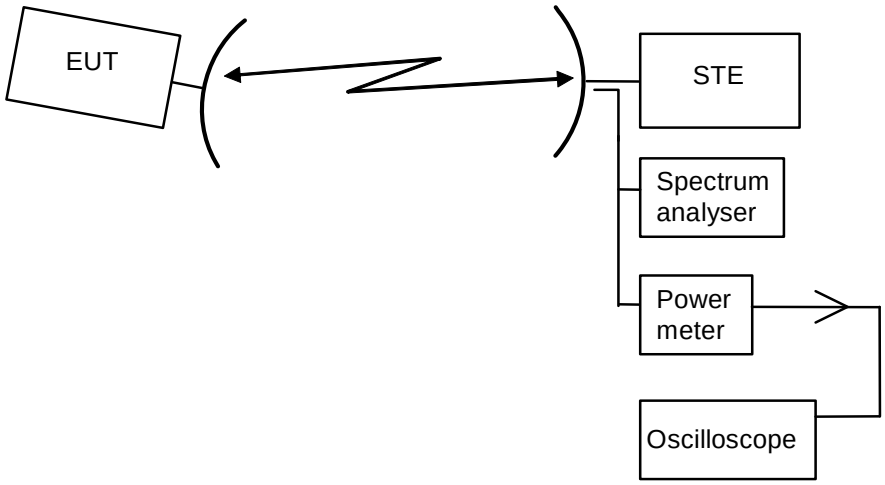


Figure 6: Test set up for pointing error detection

a) The equipment shall be tested as shown in Figure 6. An actual satellite link may be used in place of the STE. In the case of using an actual satellite link, initial step shall be taken so that interference to adjacent satellites does not occur due to the pointing errors that will be induced in the test procedure (e.g. suppress the carrier or use a low e.i.r.p.).

b) The signal received by the EUT shall be adjusted so that the signal to noise ratio corresponds to the minimum threshold signal to noise ratio defined for the intended use of the VMES for reliable tracking.

c) The spectrum analyser or the power meter shall be used to verify that the EUT has suppressed the carrier.

d) The true pointing error values defined for the intended use of the VMES shall be measured and analysed.

e) The EUT shall be capable to dynamically induce pointing errors that exceed the threshold defined for the intended use of the VMES. The induced pointing errors shall be representative of operating vehicle dynamics, within the dynamic limits for testing the EUT, as defined for the intended use of the VMES in clause 6.7.1. Pointing errors shall be induced in at least two orthogonal directions; but not necessarily during the same test sequence.

f) Conduct of test:

   1) EUT shall be switched on and pointing towards the STE in a stable tracking state.

   2) Pointing errors of an increasing magnitude shall be induced in the EUT using the test set up e) above until the defined pointing error threshold is exceeded.

   3) It shall be verified that the EUT correctly detects that the induced pointing error has exceeded the pointing error threshold and enters the "Carrier-off" radio state no later than T seconds after the pointing error threshold is exceeded, where T is the response time defined for the intended use of the VMES.

   4) Using the dynamic means given in e), the induced pointing error shall be decreased until it is less than the pointing error threshold.

   5) It shall be verified that the EUT does not re-enter the "Carrier-on" radio state for at least 2 × T seconds.

g) Step f) shall be repeated ten times.

h) If the results in f) are satisfied on all ten tests, then the EUT shall be declared compliant with the requirements.

6.7 Antenna mechanical stability

6.7.1 Test method

The methodology described in annex B shall be used to show compliance with the specification for mechanical stability.

6.8 Void

6.9 Cessation of emissions of the VMES

6.9.0 General

There shall be a means of simulating a condition where cessation of emissions is required. The spectrum analyser or the oscilloscope may be used to measure the time difference between the occurrence of the condition and the cessation of emissions. In all test methods, it shall be verified that the time difference does not exceed 1 second as specified in clause 4.2.6.2.3.

Where the EUT adds an operational margin around areas where cessation of emissions is required to account for the position determination accuracy and latency, the test procedure may simulate the actual behaviour of an EUT as its enters into an area where cessation of emissions is required. For example, if the EUT adds a margin of X metres around areas where cessation of emissions is required, then the test may begin with the EUT being simulated at a position more than X metres outside of an area where cessation of emissions is required. The test may then simulate the motion of the EUT towards the area where cessation of emissions is required. In any event, the EUT shall enter the "emission disabled" radio state within 1 second of entering the actual area where cessation of emissions is required.

For the purpose of these tests the EUT is the VMES either with, or without its antenna connected.

6.9.1 Test Method

6.9.1.1 Required documentation

The EUT documentation shall provide the evidence to verify the mode(s) of cessation of emissions implemented in the EUT and the VMES interfaces involved in the cessation of emissions as specified in clause 4.2.6.2.1.

The EUT documentation shall provide the evidence to verify compliance with specification 2 in clause 4.2.6.2.2 for the determination of the conditions under which the VMES shall cease emissions.

6.9.1.2 Cessation of emissions from the "Transmission-enabled" state

a) The EUT shall be set in the "Transmission-enabled" VMES state.

b) The EUT shall be set in the "Carrier-on" radio state.

c) A condition requiring the cessation of emission shall be initiated either by the STE or the EUT.

d) It shall be verified that the EUT enters the "Emissions-disabled" radio state.

e) In order to verify that the EUT is in the "Transmission-disabled" VMES state, it shall be verified that the EUT remains in the "Emissions-disabled" radio state when the EUT is requested to transmit data.

6.9.1.3 Cessation of emission from the "Transmission-disabled" state

a) The EUT shall be set in the "Transmission-disabled" state.

b) A condition requiring the cessation of emission shall be initiated either by the STE or the EUT.

c) In order to verify that the EUT remains in the "Transmission-disabled" VMES state, it shall be verified that the EUT remains in the "Emissions-disabled" radio state when the EUT is requested to transmit data.

d) The STE shall send a TxE command to the EUT.

e) In order to verify that the EUT remains in the "Transmission-disabled" VMES state, it shall be verified that the EUT remains in the "Emissions-disabled" radio state when the EUT is requested to transmit data.

f) The STE shall send a TxD command to the EUT.

g) The condition requiring the cessation of emissions shall be removed.

h) In order to verify that the EUT remains in the "Transmission-disabled" VMES state, it shall be verified that the EUT remains in the "Emissions-disabled" radio state when the EUT is requested to transmit data.

6.9.1.4 Cessation of emission from the "Initial Phase" state

6.9.1.4.1 EUTs transmitting initial bursts

This applies only to EUTs transmitting initial bursts (i.e. where "Initial phase-BurstOn" and "Initial phase-BurstOff" are implemented):

a) The EUT shall be set in "Initial phase-BurstOn" state.

b) The STE shall not send the TxE command and the EUT shall be allowed to cycle between "Initial phase-BurstOn" and "Initial phase-BurstOff" as it would under normal operations.

c) A condition requiring the cessation of emission shall be initiated either by the STE or the EUT.

d) It shall be verified that the EUT enters the "Emissions-disabled" radio state.

e) In order to verify that the EUT is in the "Initial phase-Standby" VMES state, it shall be verified that the EUT remains in the "Emissions-disabled" radio state when the EUT is requested to transmit data.

f) The STE shall send a TxE command to the EUT.

g) In order to verify that the EUT remains in the "Initial phase-Standby" VMES state or enters the "Transmission-disabled" VMES state, it shall be verified that the EUT remains in the "Emissions-disabled" radio state when the EUT is requested to transmit data.

h) Repeat the test steps a) through g) using a TxD command in step f) instead of the TxE command.

6.9.1.4.2 EUTs not transmitting initial bursts

This applies only to EUTs that do not transmit initial bursts (i.e. where only "Initial phase-Standby" is implemented):

a) The EUT shall be set in "Initial phase-Standby" VMES state.

b) The STE shall not send the TxE command to maintain the EUT in "Initial phase-Standby" VMES state.

c) A condition requiring the cessation of emission shall be initiated either by the STE or the EUT.

d) In order to verify that the EUT remains in the "Initial phase-Standby" VMES state, it shall be verified that the EUT remains in the "Emissions-disabled" radio state when the EUT is requested to transmit data.

e) The STE shall send a TxE command to the EUT.

f) In order to verify that the EUT remains in the "Initial phase-Standby" VMES state or enters the "Transmission-disabled" VMES state, it shall be verified that the EUT remains in the "Emissions-disabled" radio state when the EUT is requested to transmit data.

g) Repeat the test steps a) through f) using a TxD command in step e) instead of the TxE command.

6.9.1.5 "Single action" means of cessation of emissions

a) It shall be verified that there is a "single action" means of ceasing emissions of the EUT (e.g. the switch thrown).

b) The EUT shall be set in the "Transmission-enabled" VMES state.

c) The EUT shall be set in the "Carrier-on" radio state.

d) The "single action" means of cessation of emissions shall be activated.

e) It shall be verified that the EUT enters the "Emissions-disabled" radio state.

f) In order to verify that the EUT is in the "Transmission-disabled" VMES state, it shall be verified that the EUT remains in the "Emissions-disabled" radio state when the EUT is requested to transmit data.

g) The STE shall send a TxE command to the EUT.

h) In order to verify that the EUT remains in the "Transmission-disabled" VMES state, it shall be verified that the EUT remains in the "Emissions-disabled" radio state when the EUT is requested to transmit data.

i) The EUT shall be switched off and then switched on and the EUT shall enter the "Initial phase" VMES state without receiving a TxE from the STE.

j) The "single action" means of cessation of emissions shall be activated.

k) The STE shall send a TxE command to the EUT.

l) In order to verify that the EUT remains in the "Initial phase" VMES state or enters the "Transmission-disabled" VMES state, it shall be verified that the EUT remains in the "Emissions-disabled" radio state when the EUT is requested to transmit data.

m) The EUT shall be switched off and then switched on and the EUT shall enter the "Initial phase" VMES state without receiving a TxE from the STE.

n) The STE shall send a TxD command to the EUT so that it enters the "Transmission-disabled" VMES state.

o) The "single action" means of cessation of emissions shall be activated.

p) The STE shall send a TxE command to the EUT.

q) In order to verify that the EUT remains in the "Transmission-disabled" VMES state, it shall be verified that the EUT remains in the "Emissions-disabled" radio state when the EUT is requested to transmit data.

6.9.1.6 Fault conditions

The EUT documentation shall include any means of generating fault conditions.

a) The EUT shall be set in the "Transmission-enabled" VMES state.

b) The EUT shall be set in the "Carrier-on" radio state.

c) A fault condition requiring the cessation of emission shall be generated either by the STE or directly on the EUT.

d) It shall be verified that the EUT enters the "Emissions-disabled" radio state as long as the fault condition persists.

e) Steps a) through d) shall be repeated for each fault condition as defined for the intended use of the VMES.

6.10 Identification of VMES

6.10.1 Test arrangement

The test shall be arranged such that EUT can determine its own location, or if external means are employed, the test shall be arranged so that the EUT can be located by the means defined for the intended use of the VMES. The STE shall be arranged to simulate the ACF. The actual location of the EME shall be known within the resolution required in clause 4.2.7.2.

6.10.2 Test method

a) The EUT shall be in the "Transmission-enabled" state.

b) The EUT shall report its location to the STE.

c) The difference between the reported EUT location and actual location of the EUT (as simulated in the test) shall be within twice the resolution required in clause 4.2.7.2.

6.11 Control and monitoring functions

6.11.0 General

The VMES can be modified for the purpose of these tests provided that full documentation is given to prove that the modifications accurately simulate the required test conditions.

The EUT shall transmit at $EIRP_{max}$.

For the purpose of these tests the EUT is the VMES either with, or without its antenna connected.

The measurement of the e.i.r.p. spectral density shall be limited within either the nominated bandwidth or to a 10 MHz bandwidth centred on the carrier frequency, whichever is the greater.

6.11.1 Test arrangement

The EUT shall be authorized to transmit and shall be in the "Transmission-enabled" state at the commencement of each test, unless otherwise stated. The time difference between the command, or failure, and the occurrence of the expected event (e.g. the transmission suppression) shall be measured and recorded.

6.11.2 Processor monitoring-Test method

a) Each of the processors within the EUT shall, in turn, be caused to fail.

b) Within 1 second of such failure the EUT shall cease to transmit.

c) It shall be confirmed that the transmissions have been suppressed within the limits of the "Carrier-off" radio state emission levels.

d) Within 30 seconds of such failure it shall be confirmed that the transmissions have been suppressed within the limits of the "Emissions-disabled" radio state emission levels.

e) The failed processor shall be restored to normal working condition and the EUT shall restore automatically to normal working before the next processor shall be induced to fail.

6.11.3 Transmit subsystem monitoring-Test method

a) The frequency generation subsystem within the EUT shall be caused to fail in respect of:

- loss of frequency lock (if applicable);

- absence of Local Oscillator (LO) output signal.

b) Recognition of each failure in turn by the subsystem monitor shall constitute an SMF event.

c) Within 1 second of such failure the EUT shall cease to transmit.

d) It shall be observed that the transmissions have been suppressed within the limits of the "Emissions-disabled" radio state emission levels.

e) The failed elements shall be restored to normal working state and the EUT shall be restored to normal working condition before the next induced failure.

6.11.4 Power-on/Reset-Test method

a) The EUT shall be switched off and the STE shall not transmit the CC.

b) The EUT shall be switched on.

c) The EUT shall not transmit during and after switching-on, and shall enter the "Non-valid" state.

If a manual reset function is implemented the following test shall be performed:

a) The EUT shall be restored to the "Initial phase" state and the STE shall transmit the CC.

b) The EUT shall remain in the "Initial phase" state.

c) The reset function shall be initiated.

d) The EUT shall enter the "Non-valid" state.

e) The EUT shall be restored to the "Initial phase" state and the STE shall transmit the CC as well as a TxE.

f) The EUT shall enter the "Transmission-enabled" state.

g) The reset function shall be initiated.

h) The EUT shall enter the "Non-valid" state.

6.11.5 Control Channel and Response Channel-Test method

For the purposes of these test, the period without correct reception of the CC (T1) shall be as specified in clause 4.2.8.5.2.2.

These test procedures require simulation of the following events:

- The CC has never been received by the EUT after power-on.
- The CC is lost by the EUT after reception of a Transmission Enable command.
- The CC is lost by the EUT without reception of a Transmission Enable command.
- The CC is being lost by the EUT and a call is initiated within the period T1.

a) Case where the CC has never been received by the EUT after power-on:

a1) the EUT shall be switched off and the STE shall not transmit the CC;

a2) the EUT shall be switched on;

a3) the EUT shall remain in the "Non-valid" state.

b) Case where the CC is lost by the EUT after reception of a Transmission Enable command:

b1) the EUT shall be switched-on and the STE shall transmit the CC and a Transmission Enable command;

b2) the EUT shall enter the "Initial phase" state and go, if applicable, to the "Transmission-enabled" state;

b3) a transmission request shall be initiated from the EUT;

b4) the STE shall stop transmitting the CC;

b5) within the period T1 from event b4), the EUT shall enter the "Non-valid" state.

c) Case where the CC is lost by the EUT without reception of a Transmission Enable command:

c1) the EUT shall be switched on and the STE shall transmit the CC;

c2) the EUT shall enter the "Initial phase" state;

c3) the STE shall stop transmitting the CC;

c4) the EUT shall enter in the "Non-valid" state not later than T1;

c5) a transmission request shall be initiated and the EUT shall remain in the "Non-valid" state.

d) Case where the CC is being lost by the EUT and a call is initiated within the T1 period:

d1) the EUT shall be switched on and the STE shall transmit the CC;

d2) the STE shall stop transmitting the CC;

d3) within the period T1 from d2), a transmission request shall be initiated from the EUT;

d4) the EUT may transmit but within the T1 period the EUT shall enter the "Non-valid" state.

6.11.6 Network Control commands-Test method

The tests shall be performed in the following sequence:

- Transmission Enable command.
- Transmission Disable command received in the "Transmission-enabled" state.
- Transmission Disable command received in the "Initial phase" state.

a) Transmission Enable command:

a1) the EUT shall be switched-on and the STE shall transmit the CC;

a2) the EUT shall enter the "Initial phase" state;

a3) a transmission request shall be initiated from the EUT, the EUT shall remain in the "Initial phase" state;

a4) the STE shall transmit a transmit enable command to the EUT;

a5) a transmission request shall be initiated from the EUT;

a6) the EUT shall enter the "Transmission-enabled" state and shall transmit.

b) Transmission Disable command received in the "Transmission-enabled" state:

b1) continue from a6);

b2) the STE shall transmit a disable command to the EUT;

b3) the EUT shall enter the "Transmission-disabled" state within 1 second;

b4) a transmission request shall be initiated from the EUT;

b5) the EUT shall remain in the "Transmission-disabled" state;

b6) the STE shall transmit an enable command;

b7) the EUT shall enter either the "Transmission-enabled" state or the "Initial phase" state;

b8) if the EUT is in the "Transmission-enabled" state then the test continues with b11);

b9) the STE shall transmit a TxE command;

b10) the EUT shall enter the "Transmission-enabled" state;

b11) if a transmission request is not active anymore, then a new transmission request shall be initiated;

b12) the EUT shall transmit;

b13) the EUT transmission shall be terminated.

c) Transmission disable command received in the "Initial phase" state:

c1) the EUT shall be switched-on and the STE shall transmit the CC;

c2) the EUT shall enter the "Initial phase" state;

c3) the STE shall transmit a Transmission Disable command to the EUT;

c4) the EUT shall enter the "Transmission-disabled" state within 1 second;

c5) a transmission request shall be initiated from the EUT;

c6) the EUT shall remain in the "Transmission-disabled" state;

c7) the STE shall transmit a Transmission Enable command;

c8) the EUT shall enter either the "Transmission-enabled" state or the "Initial phase" state;

c9) if the EUT is in the "Transmission-enabled" state then the test continues with c12);

c10) the STE shall transmit a TxE command;

c11) the EUT shall enter the "Transmission-enabled" state;

c12) if a transmission request is not active any more then a new transmission request shall be initiated;

c13) the EUT shall transmit;

c14) the EUT transmission shall be terminated.

6.11.7 Initial burst transmission-Test method

The EUT documentation shall include any means to identify the initial burst duration.

The following test shall be performed in sequence:

a) the EUT shall be switched-off and the STE shall transmit the CC;

b) the EUT shall be switched-on;

c) the EUT shall not transmit, except the initial bursts;

d) it shall be verified that the specifications given in clause 4.2.8.7.2 are fulfilled.

6.11.8 Inhibition of transmission-Test method

The EUT documentation shall include the values for IT_{max} and IT_{trans}, as specified in clause 4.2.8.8.

The receive carrier to the EUT shall be removed:

a) Within 0,5 seconds of such failure the EUT shall cease to transmit.

b) It shall be observed that the EUT remains in the "Transmission-enabled" state and that transmissions have been suppressed within the limits of the "Carrier-off" radio state emission levels given in clause 4.2.1.2, Table 3.

c) The receive carrier to the EUT shall be restored within IT_{max}.

d) It shall be observed that the EUT enters the "Carrier-on" radio state and begins transmission within IT_{trans}.

e) The receive carrier to the EUT shall be removed.

f) It shall be observed that after IT_{max} the EUT enters the "Initial phase" state.

6.12 Receive antenna off-axis gain pattern

6.12.1 Test Method

6.12.1.1 Test site

This test shall be performed using the method of measurement and test arrangement as specified in the following clauses. The tests can be carried out in an Open Area Test Site (OATS) environment or in an anechoic chamber or in a semi-anechoic chamber as they lead to comparable test results for establishing compliance. Ambient noise levels shall be at least 6 dB below the applicable unwanted emissions limit. The preferred test site is the semi-anechoic chamber.

6.12.1.2 Method of measurement

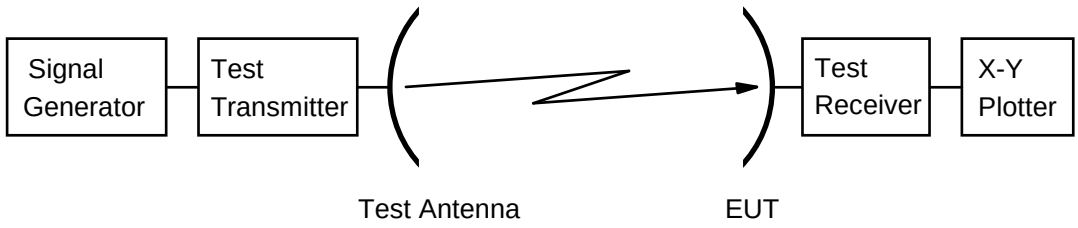


Figure 7: Test arrangement - antenna receive pattern measurement

a) The test arrangement shall be as shown in Figure 7 with the EUT connected to the test receiver.

b) A signal proportional to the angular position from the servo mechanism shall be applied to the X-axis and the signal level from the test receiver shall be applied to the Y-axis of the plotter.

c) The test frequencies shall be the centre frequency of each applicable frequency range. The E plane shall be vertical.

d) The EUT shall be aligned to maximize the received signal level and the X-Y plotter shall be adjusted to give the maximum reading on the chart.

e) The EUT shall be driven in azimuth through 180°.

f) The pattern measurement is then obtained by driving the EUT in azimuth through 360° with the plotter recording the results.

g) The tests in b) to e) shall be repeated with the frequency changed to the lower limit of the applicable band as declared by the manufacturer.

h) The tests in b) to e) shall be repeated with the frequency changed to the upper limit of the applicable band as declared by the manufacturer.

i) The tests in b) to h) shall be repeated with the frequencies changed to the others specified if the design of the equipment is such that operation is possible, but not necessarily simultaneously, in all bands.

j) The tests in b) to h) shall be repeated with the test signal being transmitted in the H-plane instead of the E-plane.

k) The tests in b) to h) shall be repeated with the test signal being transmitted in a plane at 45° to the H-plane.

l) The tests in b) to h) shall be repeated with the test signal being transmitted in a plane at 90° to that in k).

m) The tests in b) to l) shall be repeated between the angles of φ_r and 7° with the EUT rotated through 90°, or the test antenna or the polarization subsystem of the EUT rotated by 90°, to give the cross-polar measurement.

6.13 Blocking performance

6.13.1 Test method

a) The output signals of two signal generators shall be combined with equal weight. The first signal shall be the wanted modulated signal as defined for the intended use, tuned at the centre frequency of the receive frequency band and have a level in accordance with the defined intended use. The second signal shall be a CW signal with same level. The combined signal shall be coupled to the LNB input.

b) A spectrum analyser shall be connected to the LNB output in a way that allows to supply the LNB with power.

c) f_c is the centre frequency of the receive frequency band.

d) The first signal generator frequency shall be set to f_c.

e) The first signal generator level shall be set to a level in the LNB operational input level range.

f) The spectrum analyser shall be set for measuring the level of the converted first signal at the LNB output.

g) The second signal generator frequency shall be set to f_c-20 "MHz".

h) The second signal generator level shall be adjusted so that the measured level is 1 dB less than in absence of the second signal.

i) The second signal generator level shall be noted down as reference level.

j) The second signal generator frequency shall be set to the frequency of interest.

k) The second signal generator level shall be adjusted so that the measured level is 1 dB less than in absence of the second signal.

l) The rejection at the frequency of interest is equal to the second signal generator level minus the reference level determined in step i).

m) Steps j) to l) shall be repeated for frequencies in the ranges of Table 4.

NOTE: The worst case rejection in a particular frequency range can be determined after step i) by sweeping the second signal generator frequency over the frequency range and observe the gain compression, then perform steps j) to l) with the frequency where gain compression is highest.

6.14 Adjacent Signal Selectivity

6.14.1 Test method

a) Two test signal generators shall be used. Each signal generator shall generate the same modulated signal as the wanted signal in the IME input frequency range and thermal noise as defined for the intended use.

b) The signal generators shall be connected to the IME input through a splitter (combiner).

c) The test signal generators shall be set to the frequencies and levels according to Table 5.

d) The IME shall be set to receive the signal of the first test signal generator.

e) The second test signal generator shall be set to signal off.

f) The noise level (or signal to noise ratio) of the first test signal generator shall be varied in order to determine the threshold reference sensitivity level, such that the SNR (at the receiver input) is set to the reference level plus 1 dB. The reference level is the minimum SNR required to achieve the target performance level for the modulated wanted signal. This is the sensitivity reference level.

g) The second signal generator shall be set to signal on.

h) The noise level (or signal to noise ratio) of the first test signal generator shall be varied in order to determine the threshold reference sensitivity level.

i) The degradation is equal to the noise level (or signal to noise ratio) determined in step h minus that determined in step f.

j) The result is the highest degradation found.

Annex A (informative): Relationship between the present document and the essential requirements of Directive 2014/53/EU

The present document has been prepared under the Commission's standardisation request C(2015) 5376 final [i.5] to provide one voluntary means of conforming to the essential requirements of Directive 2014/53/EU on the harmonisation of the laws of the Member States relating to the making available on the market of radio equipment and repealing Directive 1999/5/EC [i.6].

Once the present document is cited in the Official Journal of the European Union under that Directive, compliance with the normative clauses of the present document given in Table A.1 confers, within the limits of the scope of the present document, a presumption of conformity with the corresponding essential requirements of that Directive, and associated EFTA regulations.

Table A.1: Relationship between the present document and the essential requirements of Directive 2014/53/EU

Harmonised Standard ETSI EN 302 977					
Requirement				Requirement Conditionality	
No	Description	Essential requirements of Directive	Clause(s) of the present document	U/C	Condition
1	Off-axis spurious radiation	3.2	4.2.1	U	
2	On-axis emissions	3.2	4.2.2	U	
3	Off-axis e.i.r.p. emission density within the band	3.2	4.2.3	U	
4	Carrier suppression	3.2	4.2.4	U	
5	Antenna pointing	3.2	4.2.5	U	
6	Cessation of emissions of the VMES	3.2	4.2.6	U	
7	Identification of VMES	3.2	4.2.7	U	
8	Control and Monitoring Functions (CMFs)	3.2	4.2.8	U	
9	Antenna mechanical stability	3.2	4.2.5	U	
10	Antenna pointing error	3.2	4.2.5	U	
11	Receive antenna off-axis gain pattern	3.2	4.2.9	C	Satellite communications networks may contain a central control unit that keeps received interference low by taking into account the actual antenna gain pattern of earth stations. The requirement is not relevant for earth stations that are always operated as part of such a network. The requirement is relevant in all other cases. If the receive antenna performance does not meet the requirement of clause 4.2.9, then the earth station accepts the risk of additional co-frequency interference caused by its off-axis gain that is not compliant with the mask.
12	Blocking performance	3.2	4.2.10	U	
13	Adjacent Signal Selectivity	3.2	4.2.11	U	

Harmonised Standard ETSI EN 302 977					
Requirement				Requirement Conditionality	
No	**Description**	**Essential requirements of Directive**	**Clause(s) of the present document**	**U/C**	**Condition**
14	Receiver radio-frequency intermodulation	3.2			In an FSS network, receiver radio-frequency intermodulation is not relevant, because possibly inter-modulating signals originating from other networks are rejected thanks to requirements on receive antenna off-axis gain pattern and adjacent signal selectivity. Interfering signals up to 28 dB above the wanted carrier can be tolerated with minimal impact on performance.
15	Receiver sensitivity	3.2			There is no causal relationship between receiver sensitivity and interference in the case of satellite communications. Therefore, a quantitative calculation is not possible.
16	Receiver spurious response rejection	3.2			Frequencies other than the image frequency are not relevant because this is covered by receiver blocking and adjacent channel requirements which are already included in the receiver parameters.
17	Receiver multiple signal selectivity - receiver adjacent signal selectivity	3.2			The proper function of the wanted signal in the presence of other unwanted emissions is controlled via proper assignment of frequencies and frequency coordination procedures as applicable. Due to the operation without protection against interference, there is no need of specifying this parameter.
18	Receiver dynamic range	3.2			In an FSS network, receiver dynamic range has no influence on harmful interference. The satellite earth stations are deployed in a manner that results in a situation where it is impossible to receive a wanted signal that is high enough to produce any overloading effect (satellite systems are power limited). So that it is not necessary to specify receiver dynamic range. See clause 5.3.6.1 of ETSI EG 203 336 [i.8].
19	Reciprocal mixing	3.2			For FSS equipment, the parameter is excluded for the following reason. It is considered that the reciprocal mixing effects are implicitly covered in HSs where comprehensive interference characteristics are specified in terms of selectivity and/or blocking requirements, thus removing the need for this parameter to be included in HSs as the effects of receiver selectivity and reciprocal mixing cannot be separated. See clause 5.3.6.2 of ETSI EG 203 336 [i.8].

Harmonised Standard ETSI EN 302 977					
Requirement				Requirement Conditionality	
No	Description	Essential requirements of Directive	Clause(s) of the present document	U/C	Condition
20	Desensitization	3.2			As desensitization is a receiver effect addressed by other parameters (including receiver blocking), its inclusion as a separate parameter in an HS is not required. See clause 5.3.2.3 of ETSI EG 203 336 [i.8].
21	Receiver unwanted emissions in the spurious domain	3.2			Frequencies other than the image frequency are not relevant because this is covered by receiver blocking and adjacent channel requirements which are already included in the receiver parameters.

Key to columns:

Requirement:

No A unique identifier for one row of the table which may be used to identify a requirement.

Description A textual reference to the requirement.

Essential requirements of Directive

Identification of article(s) defining the requirement in the Directive.

Clause(s) of the present document

Identification of clause(s) defining the requirement in the present document unless another document is referenced explicitly.

Requirement Conditionality:

U/C Indicates whether the requirement is unconditionally applicable (U) or is conditional upon the manufacturer's claimed functionality of the equipment (C).

Condition Explains the conditions when the requirement is or is not applicable for a requirement which is classified "conditional".

Presumption of conformity stays valid only as long as a reference to the present document is maintained in the list published in the Official Journal of the European Union. Users of the present document should consult frequently the latest list published in the Official Journal of the European Union.

Other Union legislation may be applicable to the product(s) falling within the scope of the present document.

Annex B (normative): Mechanical stability methodology

This test based on an analytical or numerical analysis is performed in two stages:

- In the first stage the loads generated by the maximum wind speed is computed on the EME, taking into account the intrinsic properties of the materials.
- In the second stage the computed loads is applied to the structure.

The purpose of the analysis is twofold:

a) To show that the torque and the fields of force applied to the EME structure under nominated conditions do not reach any of the following:

- the point where deformation fouls the antenna pointing mechanism;
- the yield point of any element of the structure;
- the buckling point of any element of the structure.

NOTE: The yield point is elastic limit of the material and the buckling point is the failure point for the material.

b) To compute equivalent static loads (force and torque) applied to the critical attachment points of the structures, e.g.:

- for antennas without a radome:
  - reflector-mounting legs fixing point;
  - reflector-struts;
  - LNB- struts;
- for antennas with a radome:
  - radome surface;
  - radome mounting points.

Analysis and load applications procedure:

a) The air related parameters, namely the kinetic viscosity used to calculate drags at the rims of the structure is calculated with the standard atmospheric environmental conditions (temperature = 293 K, air pressure = $1{,}013 \times 10^5$ Pascal).

b) The computations needed to derive the field of force and torque and the equivalent static stresses is carried out for each of the following variables:

- elevation angle: maximum and minimum for antennas without a radome;
- wind direction: in steps of 45° around the EME unit;
- maximum relative wind speed: 100 km/h;
- gust wind speed: 130 km/h for 3 seconds.

Verification of the simulated results shall be carried out to ensure that the yield point limits were not exceeded for any self-contained element.

The calculated equivalent static loads is applied at any identified critical fixing point of the assembly, whilst the loads are applied the above deck unit is observed and any distortion noted.

The test report shall contain the following information:

- the computation method used;
- the description of the test equipment;
- the description of the tests performed;
- the results of the safety margin test;
- any signs of distortion observed;
- the results of the measurements of the deviation of the antenna or radome position;
- the component deviation with respect to each other.

Annex C (informative): Typical Scan and Acceleration Rates

Table C.1 indicates the typical antenna scan and acceleration rates required to maintain performance under the environmental profile of clause 4.1.1 in compliance with the technical specifications given in the remainder of clause 4.

Table C.1: Typical Scan and Acceleration Rates

		Class A	Class B
Scan rate	Azimuth	24°/s	400°/s
	Elevation	12°/s	400°/s
	Polarization	12°/s	400°/s
Acceleration	Azimuth	50°/s^2	500°/s^2
	Elevation	50°/s^2	500°/s^2
	Polarization	40°/s^2	400°/s^2

Annex D (informative): Maximum measurement uncertainty

The measurements described in the present document are based on the following assumptions:

- the measured value related to the corresponding limit is used to decide whether an equipment meets the requirements of the present document;
- the value of the measurement uncertainty for the measurement of each parameter is included in the test report.

Table D.1 and Table D.2 show the recommended values for the maximum measurement uncertainty figures.

Table D.1: Maximum measurement uncertainty

Parameter	Uncertainty
Radio frequency	±10 kHz
RF power	±0,75 dB
Conducted spurious	±4 dB
Radiated spurious	±6 dB
Antenna on-axis gain	±2 dB

Table D.2: Maximum measurement uncertainties for antenna gain pattern

Gain relative to the antenna on-axis gain	Uncertainty
> -3 dB	±0,3 dB
-3 dB to -20 dB	±1,0 dB
-20 dB to -30 dB	±2,0 dB
-30 dB to -40 dB	±3,0 dB

Annex E (informative): Bibliography

- ETSI ETR 169 (1995): "Satellite Earth Stations and Systems (SES); Common Technical Regulations (CTRs) in the satellite earth station equipment field".
- EN 60068: "Environmental testing. Test methods for vibration and shock", produced by CENELEC.
- EN 55022: "Limits and methods of measurement of radio disturbance characteristics of information technology equipment", produced by CENELEC.
- SAE J1211: "Recommended Environmental Practices for Electronic Equipment Design, Recommended Practice".
- CEPT/ERC/Recommendation 74-01: "Unwanted emissions in the spurious domain", Approved 1998, Amended 29 May 2019, Updated 1 October 2021.
- ETSI EG 201 399: "Electromagnetic compatibility and Radio spectrum Matters (ERM); A guide to the production of Harmonized Standards for application under the Radio & Telecommunication Terminal Equipment Directive 1999/5/EC (R&TTE) and a first guide on the impact of the Radio Equipment Directive 2014/53/EU (RED) on Harmonized Standards".

Annex F (informative): Change history

Version	Information about changes
V2.1.2	Revisions by SES HARM#83, #84 and #85. Further reconciliation of V1.1.9 approved by TC SES and previous published version V2.1.1
V2.1.3	Revisions by SES HARM#88
V2.1.4	Clean version after review by SES HARM#88
V2.1.5	Further revision at the light of ETSI secretariat review and comments
V2.1.6	Final revision by SES HARM#93 producing track changes and clean versions
V2.17 - v2.1.8	Post edits for review by TC SES
V2.1.9	Review by WG HARM 137 addressing HAS comments
V2.1.10	Continue review by WG HARM 138 addressing HAS comments
V2.1.11	Review by WG HARM 139 completing address of HAS comments
V2.1.12	Review by WG HARM 148 updating reference to OATS and preferred test site

History

Version	Date	Status
V1.1.2	February 2010	Publication
V2.1.1	June 2016	Publication
V2.1.13	August 2026	SRdAP process EV 20261119: 2026-08-27 to 2026-11-19